Reciprocal absorption–scattering asymmetry: sharp channel bounds, exact vector dielectric constructions, and a restricted thickness law

OQ7 research continuation

Analytical, numerical, and formal verification report

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Abstract

We distinguish several precise formulations of Open Question 7 in the Miller–Monticone tutorial. With all channels included, unit incident power, and a specified lossless reference, the common extinction X of reversed incident fields bounds their absorption difference by $X - X^2/4$. This is a direct consequence of established passivity inequalities. A reciprocal three-port attains the bound, whereas scalar two-ports have a smaller sharp envelope and extinction-normalized contrast at most $\sqrt{10\sqrt{5} - 22}$. With prescribed overlap between an input and its time reverse, we prove a sharper three-branch joint envelope using an explicit positive-matrix certificate; its priority remains unestablished. We give exact two-layer, nonmagnetic anisotropic dielectric realizations approaching the multichannel envelope, and extend the construction to finite-power beams with exact prescribed extinction. One reversed plane wave has exactly zero absorption, demonstrating why a positive total tensor-loss budget does not imply the scalar determinant floor. These polarization mechanisms have clear prior precedents; their novelty is not claimed. We preserve the earlier scalar-multilayer inverse-square thickness theorem, its full ordinary proof, and its numerical record. A new Lean/Mathlib project verifies a precisely enumerated subset of the algebra, finite-channel results, and the general joint-bound certificate; the physical thickness theorem remains unverified in Lean. The reported 692-theorem project and declaration `OQ7.theorem51` were absent from the available source archive. Compact-body three-dimensional plane-wave cross sections, coercive-loss resource bounds, and finite bandwidth remain unresolved.

Verification and priority conventions. **KNOWN** means established or a direct consequence of earlier results. **PROVED** refers to the complete ordinary proofs in this manuscript, independently reviewed as documented below; it does not imply formal verification. **LEAN-VERIFIED** is reserved for the exact declarations listed in the source-correspondence appendix. **CONDITIONAL** concerns the priority of the restricted material-resource and joint-overlap theorems, not a certificate of novelty. The original manuscript and numerical files are preserved without replacement.

1 Novelty check before theorem development

The starting question is OQ7 in Section 4.1 of Miller and Monticone's tutorial [1]. A two-port specialization does not by itself solve the general optical cross-section question. In Sections 2–5, absorption and transmission are power fractions for a pair of power-normalized ports. For reciprocal two-port systems,

$$A_L - A_R = |r_R|^2 - |r_L|^2.$$

Thus the same difference measures the opposite imbalance in reflected power. For an extended free-space object, identifying all scattered power with one reflection channel would require additional assumptions.

1.1 Mandatory comparisons

Prior result	Statement and assumptions	Consequence for this investigation
Sounas–Alù (2014) [2]	Reciprocity and the optical theorem give $\sigma_{\text{ext}}(\mathbf{e},\mathbf{k}) = \sigma_{\text{ext}}(\mathbf{e}^*,-\mathbf{k})$. The polarization is conjugated when propagation reverses.	Extinction symmetry is established. It permits absorption and scattering to exchange their directional contributions; it does not specify a material and thickness budget.
Sounas–Alù (2017) [3]	For a lossless time-reversal-symmetric two-port, the pointwise ratio of oppositely excited internal intensities lies between $(1-\lvert r\rvert)/(1+\lvert r\rvert)$ and its reciprocal. A multiport radiation-efficiency inequality incorporates loss channels.	Internal-field asymmetry constraints are established. Applying the lossless two-port formula unchanged to an arbitrary lossy structure is invalid.
Guo–Fan (2022) [4]	Necessary and sufficient reflection feasibility at prescribed singular values, Eqs. (24)–(25). For ordered reflection magnitudes $p \geq q$ and singular values $s_1 \geq s_2$: $p-q \leq s_1-s_2 \leq p+q \leq s_1+s_2$.	Together with $s_1 \leq 1$ and the Frobenius norm, this already solves the initial matrix problem. Its explicit slice is not a discovery.
Máñez-Espina et al. (2026) [5, 6]	Two-mode temporal coupled-mode theory for reciprocal bianisotropic metasurfaces. In the balanced, equal-frequency specialization, the precursor's Eq. (18) gives $\Delta A(\omega_\pm) = \mp r_b g^2/(\gamma^2+g^2)$, for positive coupling convention.	r_b is a background reflection amplitude, not the total response's reflection or transmitted power. This model-specific bound is not an all-profile finite-dielectric optimum.

1.2 Additional results that restrict any novelty claim

Rollett's two-port invariant [7], in reciprocal scattering notation, is

$$K_{\text{R}} = \frac{1-|r_L|^2-|r_R|^2+|\det S|^2}{2|t|^2} = 1+\frac{\det(I-S^\dagger S)}{2|t|^2}. \tag{1}$$

Consequently, the quantity $c = \det(I-S^\dagger S)/|t|^2$, including its invariance under independent lossless coatings, is established. We do not extend this invariance to arbitrary feedback embeddings.

Ma, Pestourie and Johnson [8], Eq. (35), already derive the weak-loss relation

$$\frac{|A_L-A_R|}{A_L+A_R} \leq |r_{\text{lossless}}| + O(\text{loss}).$$

Their perturbative hypothesis differs from the exact finite-loss constraints used below. Guo and Fan [9] provide complete abstract eigenvalue feasibility, and Li et al. [10] characterize coherent control for a given system. Kuang, Zhang and Miller [11] already establish material-dependent single-frequency absorption and thickness limits. Neither positivity of loss nor the general idea of a material-dependent thickness limit can be claimed as new.

Weighted observability has further close precedents. Chitour and Sigalotti [12], Lemma 2.5, prove positive Gramian floors for measurable observation of fixed oscillators. Hante, Sigalotti and Tucsnak [13], Proposition 5.4, establish quantitative small-window observability for a fixed

skew-symmetric generator and controllable observation. More explicitly, Privat, Trélat and Zuazua [14], Remark 2, give, for every measurable $\omega \subset [0,\pi]$ with $|\omega| = \rho\pi$,

$$\frac{\rho\pi - \sin(\rho\pi)}{2} \le \inf_{j\ge 1} \int_\omega \sin^2(jx)\,\mathrm{d}x \le \frac{\rho\pi + \sin(\rho\pi)}{2}.$$

This concerns worst-mode observation for a fixed homogeneous real wave operator. These results preempt a claim that the Gramian method or the local sinusoidal optimization is itself new. The spatial complex Helmholtz generator below is variable and has an indefinite flux form; identifying it with a fixed dissipative time generator does not directly transfer those theorems.

Likewise, the optical Borrmann effect already places thin absorbing layers near standing-wave minima [15, 16]. The construction below uses that established physical mechanism. Its claimed research content is restricted to a uniform all-profile bound and a matching *exact-target* leading coefficient, not a new multilayer architecture.

Priority verdict. No equivalent statement of Theorem 5.1 was found in the retrieved optics, two-port, transmission-bound, or observability sources. This is a precise candidate contribution, but the search does not certify priority. Its classification remains **CONDITIONAL**. The original search cutoff was 9 October 2026; the targeted generalization review extends it through 10 October 2026. No coverage of the unelapsed remainder of October is claimed.

The 2014 main theorem and proof were inspected in the publisher-provided first-page preview; the remaining pages were not retrieved. For the 2026 metasurface paper, published prose and the complete arXiv precursor were read; the publisher PDF and supplement were blocked. Older complex-potential bounds of Martin [17] were identified, but complete primary-text retrieval remained incomplete. These are explicit limitations of the novelty audit, rather than grounds to infer absence of prior work.

1.3 Additional priority checks for the generalization

The channel inequality in Section 6 follows from the optical-theorem passivity disk underlying absorption–scattering tradeoffs, including Fleury, Soric and Alù [18]. Its two-illumination form is not claimed as an independent discovery. The sharper scalar two-port optimizations are consequences of Guo–Fan reflection feasibility, rather than new matrix constraints.

Polarization conversion is also an established source of directional power asymmetry. Armitage [19], Eq. (27), gives reversed Jones transmission as the transpose; the example around Eq. (34) explicitly combines a polarizer with a rotated wave plate. Potton's earlier review [20], Section 5, treats reciprocal polarizing stacks. Our exact Maxwell construction supplies finite tensor coefficients, interface matching, a limiting sequence, and fixed-extinction sharpness. It is a rigorous synthesis of known mechanisms. The targeted search did not locate the identical constructive theorem, but that does not establish novelty or warrant labeling it NEW.

Guo and Fan's multimode work [9] completely characterizes the joint eigenvalues of transmission, reflection and absorption for a specified input side through Horn inequalities. It does not impose reciprocity on the full two-sided system and explicitly identifies reciprocal refinements as further work. It should neither be rediscovered nor represented as solving all reversed-pair Maxwell questions. A source-by-source comparison, including the limitations of full-text retrieval for Sounas–Alù (2014), Potton (2004), and Martin (1962), is supplied in the reproducibility archive.

2 The established two-port optimization

The optimization in this section is an established consequence of passive scattering-matrix feasibility. In particular, Guo and Fan's necessary and sufficient reflection conditions at prescribed

singular values [4] imply it by elimination. We supply an elementary proof, including phases and degeneracies, to specify the benchmark precisely; none of the results in this section is claimed as new.

Let the two ports be power normalized and write

$$S = \begin{pmatrix} r_L & t \\ t & r_R \end{pmatrix}, \qquad \mathcal{A} = I - S^\dagger S \succeq 0, \qquad T = |t|^2. \tag{2}$$

Define

$$\begin{gathered} A_L = 1 - T - |r_L|^2, \quad A_R = 1 - T - |r_R|^2, \\ a = \frac{A_L + A_R}{2}, \quad \delta = \frac{A_L - A_R}{2}, \quad x = |\delta|, \quad b = 1 - T - a. \end{gathered} \tag{3}$$

Thus $A_{L,R} = a \pm \delta$ and $|r_{L,R}|^2 = b \mp \delta$.

Proposition 2.1 (Established feasibility conditions)**.** *Real numbers (T, a, δ) are realized by a symmetric contraction if and only if*

$$0 \le T \le 1, \qquad 0 \le a \le 1 - T, \qquad 0 \le x \le \min(a, b), \qquad F(x) \ge 0, \tag{4}$$

where

$$F(x) = a^2 - x^2 - 2T\big(b - \sqrt{b^2 - x^2}\big). \tag{5}$$

Every pair $0 \le T \le 1$, $0 \le a \le 1 - T$ is feasible. For prescribed nonzero t, r_L, r_R, set $\theta = \arg r_L + \arg r_R - 2\arg t$. In addition to the first three conditions in (4), the exact phase condition is

$$\cos\theta \le \frac{a^2 - x^2 - 2Tb}{2T\sqrt{b^2 - x^2}}. \tag{6}$$

If $T|r_L r_R| = 0$, no phase condition is imposed; the remaining determinant condition is $a^2 - x^2 - 2Tb \ge 0$.

Proof. Put $z = \overline{r_L} t + \bar{t} r_R$. Direct multiplication gives

$$\mathcal{A} = \begin{pmatrix} a + \delta & -z \\ -\bar{z} & a - \delta \end{pmatrix}, \qquad \det \mathcal{A} = a^2 - x^2 - |z|^2. \tag{7}$$

A Hermitian two-by-two matrix is positive semidefinite exactly when its diagonal entries and determinant are nonnegative. The reflection magnitudes and diagonal entries give the first three conditions in (4). When all entries are nonzero,

$$|z|^2 = 2T\big(b + \sqrt{b^2 - x^2}\cos\theta\big),$$

whose minimum is $2T(b - \sqrt{b^2 - x^2})$. This proves necessity, the phase criterion, and its stated degenerate version. Conversely, the real symmetric matrix

$$S_0 = \begin{pmatrix} \sqrt{b - \delta} & \sqrt{T} \\ \sqrt{T} & -\sqrt{b + \delta} \end{pmatrix} \tag{8}$$

has nonnegative absorption diagonal and determinant $F(x)$, proving sufficiency. Taking $\delta = 0$ gives $S_0^\dagger S_0 = (1 - a)I$, proving the assertion about feasible pairs. □

Proposition 2.2 (Established sharp envelope)**.** *For a feasible pair (T, a), put $a_c(T) = (1 - T^2)/2$. Then*

$$\max |A_L - A_R| = \begin{cases} 2\sqrt{b^2 - \big(\sqrt{1 - 2a} - T\big)^2}, & 0 \le a < a_c(T), \\ 2b, & a_c(T) \le a \le 1 - T. \end{cases} \tag{9}$$

The formulas agree at the interface and include $T = 0$ and $T = 1$.

Proof. Writing $y = \sqrt{b^2 - x^2}$ yields the identity

$$F(x) = (y + T)^2 - (1 - 2a). \tag{10}$$

If $a \geq a_c(T)$, then $1 - 2a \leq T^2$ and $b \leq a$. Consequently $x = b$ is feasible and is maximal. If $a < a_c(T)$, put $y_0 = \sqrt{1 - 2a} - T > 0$. Equation (10) is nonnegative exactly when $y \geq y_0$. Since $F(0) = a^2 \geq 0$, we have $b \geq y_0$ and may set $x_* = \sqrt{b^2 - y_0^2}$. The equality $F(x_*) = 0$ implies $a^2 - x_*^2 = 2T(b - y_0) \geq 0$; hence $x_* \leq a$. The matrix (8) attains this largest feasible x. □

Complete equality characterization. Port interchange is included by allowing either sign of δ below; phases of zero entries are never assigned.

1. If $T = 0$, every optimizer is diagonal and satisfies

$$|\delta| = \min(a, 1 - a), \qquad |r_L| = \sqrt{1 - a - \delta}, \qquad |r_R| = \sqrt{1 - a + \delta}.$$

   All phases of nonzero reflections are arbitrary. For $a \leq 1/2$ one absorptance vanishes; for $a \geq 1/2$ one reflection vanishes. In particular, $a = 0$ permits arbitrary unit-modulus diagonal entries, $a = 1/2$ gives one zero and one unit-modulus reflection, and $a = 1$ gives $S = 0$.

2. If $T > 0$ and $0 \leq a < a_c(T)$, every optimizer has $\delta = \pm x_*$, the magnitudes in (8), and

$$\arg r_L + \arg r_R - 2 \arg t = \pi \pmod{2\pi}. \tag{11}$$

   All three entries are nonzero. Necessity follows because the largest possible absorption determinant at x_* is already $F(x_*) = 0$; therefore the minimizing phase is mandatory. These conditions are also sufficient. Equivalently, all optimizers are DS_0D, with D diagonal unitary. Their spectra satisfy

$$\operatorname{spec} \mathcal{A} = \{0, 2a\}, \qquad \sigma(S) = \{1, \sqrt{1 - 2a}\}. \tag{12}$$

   For $a > 0$ there is exactly one lossless coherent input direction. For $a = 0$, the maximum is zero, $\mathcal{A} = 0$, and S is unitary; the same phase condition still applies.

3. If $T > 0$ and $a_c(T) \leq a \leq 1 - T$, every optimizer has $|\delta| = b$, so one reflection is zero and the other has magnitude $\sqrt{2b}$. All phases of nonzero entries are arbitrary, since

$$\det \mathcal{A} = 2a - 1 + T^2 \geq 0. \tag{13}$$

   At $a = a_c(T)$ and $T < 1$, the singular values are $1, T$; above that interface $\mathcal{A}$ is positive definite. When $b = 0$, both reflections vanish and only the phase of t is free. The endpoint $T = 1, a = 0$ is therefore a perfect lossless transmitter with arbitrary transmission phase.

Explicit elimination from Guo–Fan. For ordered reflection magnitudes $p \geq q \geq 0$ and singular values $\sigma_1 \geq \sigma_2 \geq 0$, Eqs. (24)–(25) of [4] specialize to

$$p - q \leq \sigma_1 - \sigma_2 \leq p + q \leq \sigma_1 + \sigma_2. \tag{14}$$

The Frobenius identity fixes $\sigma_1^2 + \sigma_2^2 = p^2 + q^2 + 2T$. Put $d = \sigma_1 - \sigma_2$, $c = \sigma_1 + \sigma_2$. At this fixed sum of squares, $(c + d)/2 = \sigma_1$ is nondecreasing in d on $0 \leq d \leq c$. Its smallest allowed value occurs at $d = p - q$, with $c = \sqrt{(p + q)^2 + 4T}$, which satisfies every inequality in (14). Thus

$$\min \sigma_1 = \frac{p - q + \sqrt{(p + q)^2 + 4T}}{2}. \tag{15}$$

Requiring this norm to be at most one is equivalent to

$$0 \le q \le p \le 1, \qquad 0 \le T \le (1-p)(1+q). \tag{16}$$

Substituting $p = \sqrt{b+x}$ and $q = \sqrt{b-x}$ recovers (4) and its sharp fixed-mean slice (9). Thus the matrix optimization is directly implied by previously complete reflection feasibility, independently of whether its particular piecewise parametrization was written explicitly before.

3 The finite-dielectric problem

Fix a real vacuum wavenumber $k > 0$. Use time dependence $e^{-i\omega t}$, scalar relative permittivity $\varepsilon(z) = p(z) + iq(z)$, relative permeability one, normal incidence, and identical vacuum exterior media. A finite multilayer consists of finitely many homogeneous layers with finite coefficients and nonnegative thicknesses; zero-thickness layers may be discarded. The electric field satisfies the exact equation

$$E'' + k^2 \varepsilon(z) E = 0. \tag{17}$$

Electric field and its derivative are continuous at interfaces. No effective-medium, weak-loss, resonant-mode, or thin-film approximation is used in the lower bounds.

For the sharp theorem impose

$$1 \le p(z) \le P, \qquad 0 \le q(z) \le Q, \qquad \int q(z)\,\mathrm{d}z = M, \quad P > 1, \quad Q, M > 0. \tag{18}$$

The total geometric thickness includes lossless spacers and matching layers. There is no prescribed layer count or positive minimum feature size. Here M has dimensions of length and is an integrated imaginary permittivity at the specified frequency; it is not absorbed power or a frequency-integrated material sum rule. The class allows continuously selectable permittivities within the stated caps, rather than a fixed catalogue of measured materials.

3.1 Well-defined scattering and the Gram identity

The outgoing homogeneous problem has no nonzero solution at real k: outward radiated flux plus nonnegative volume dissipation would otherwise sum to zero with a positive term. Finite transfer matrices therefore give a well-defined reciprocal passive scattering matrix. Its transmission amplitude is nonzero. Indeed, zero transmission for a left incident wave would make both E and E' zero at the right boundary, so ODE uniqueness would make the entire field zero, contradicting the unit incident wave.

Let E_L, E_R denote the fields for unit incidence from the left and right. With $v(z) = (E_L(z), E_R(z))$, flux balance gives

$$\mathcal{A} = I - S^\dagger S = k \int q(z) v(z)^\dagger v(z)\,\mathrm{d}z. \tag{19}$$

For any subinterval J, denote the same integral restricted to J by $\mathcal{A}_J$. It is positive semidefinite and $\mathcal{A} - \mathcal{A}_J \succeq 0$.

Let $s(y, x)$ be the fundamental solution of (17) with $s(x, x) = 0$ and $\partial_y s(x, x) = 1$. Constancy of the Wronskian gives

$$E_L(x) E_R'(x) - E_L'(x) E_R(x) = -2ikt,$$

and hence

$$E_L(x) E_R(y) - E_R(x) E_L(y) = -2ikt\, s(y, x).$$

Expanding the determinant of the two-dimensional Gram integral now proves the exact identity

$$\boxed{\frac{\det \mathcal{A}_J}{T} = 4k^4 \int_{\substack{x<y \\ x,y\in J}} q(x)q(y)|s(y,x)|^2 \,\mathrm{d}x\,\mathrm{d}y.} \tag{20}$$

The denominator is the transmission of the complete stack, including surroundings. In particular, positive loss on any interval makes $\mathcal{A}$ positive definite: a zero quadratic form in (19) would force a nonzero incident combination to vanish on that interval, contradicting uniqueness. Thus no finite stack with positive distributed loss attains the rank-one active-branch matrix optimum. This qualitative observation alone gives no uniform finite-thickness gap.

3.2 Transfer form and the established coating invariant

For the field state $u = (E, E'/(ik))^{\mathsf{T}}$, let F transfer the left boundary to the right boundary. It has determinant one. Put $\Sigma = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$. The normalized directed flux is $u^\dagger \Sigma u/2$. The matrix mapping incident amplitudes to the left field state is

$$B_{\mathrm{in}} = \begin{pmatrix} 1 + r_L & t \\ 1 - r_L & -t \end{pmatrix}, \qquad \det B_{\mathrm{in}} = -2t.$$

Consequently

$$\mathcal{A} = \tfrac{1}{2} B_{\mathrm{in}}^\dagger (\Sigma - F^\dagger \Sigma F) B_{\mathrm{in}},$$

and

$$c := \frac{\det \mathcal{A}}{T} = \det(\Sigma - F^\dagger \Sigma F). \tag{21}$$

If F is replaced by $F_R F F_L$, with $F_j^\dagger \Sigma F_j = \Sigma$ and $\det F_j = 1$, the determinant is unchanged. This directly verifies the coating invariance used below, while (1) identifies its classical provenance.

4 A uniform material-resource lower bound

For this section it suffices to have $p \le P$, $0 \le q \le Q$ on a selected interval J of length $L > 0$, with $P, Q > 0$ and $\int_J q = M > 0$. Additional finite passive layers outside J are allowed. Define

$$K = k\sqrt{P}, \qquad f(u) = u^2 - \sin^2 u, \qquad g(u) = \sqrt{f(u)}. \tag{22}$$

Lemma 4.1 (Complex modulus comparison)**.** *For $0 \le h \le \pi/K$ with $[x, x+h] \subseteq J$,*

$$|s(x+h, x)| \ge \frac{\sin(Kh)}{K}. \tag{23}$$

Proof. A zero at distance $h < \pi/K$, together with integration by parts and real parts of (17), would imply

$$\int_x^{x+h} |s'|^2 = k^2 \int_x^{x+h} p|s|^2 \le K^2 \int_x^{x+h} |s|^2.$$

Dirichlet Poincaré gives the incompatible lower bound $(\pi/h)^2 \int |s|^2$. Thus $\rho = |s|$ is positive before π/K. Writing $j = \mathrm{Im}(\bar{s}s')$, direct differentiation away from the initial zero gives

$$\rho'' + K^2 \rho = (K^2 - k^2 p)\rho + \frac{j^2}{\rho^3} \ge 0, \qquad \rho(0) = 0, \quad \rho'(0) = 1.$$

Near the origin $\rho(h) = h + O(h^3)$, so the initial conditions and the integrability needed for Duhamel's formula are valid. The kernel $\sin(K(h-v))/K$ is nonnegative for $0 \le v \le h \le \pi/K$. Duhamel's formula proves the claim on the open interval, and continuity gives the endpoint. Interface continuity justifies the same argument for piecewise constant coefficients. □

Lemma 4.2 (One-period capped Fourier moment)**.** *If $0 \le q \le Q$ on an interval of length at most π/K, and $m = \int q$, then*

$$\left| \int q(z) e^{2iKz} \, \mathrm{d}z \right| \le \frac{Q}{K} \sin\!\left(\frac{Km}{Q} \right). \tag{24}$$

Proof. Extend q by zero to one period and set $\theta = 2Kz$. For a chosen phase ϕ, maximizing $\int q \cos(\theta - \phi) \, \mathrm{d}\theta$ at fixed mass fills the arc where the cosine is largest at density Q. Indeed, if q_* is this saturated arc and c_* is its endpoint cosine, then $(q_* - q)(\cos(\theta - \phi) - c_*) \ge 0$ pointwise, and the masses of q_*, q agree. Integrating proves optimality. The arc has angular half-width Km/Q; its moment is $Q \sin(Km/Q)/K$. Choose ϕ to align with the original complex moment. The arc may wrap across the period boundary; allowing this only enlarges the feasible class. This standard mass-allocation argument is included to fix constants, not as a claim of a new moment inequality. □

Lemma 4.3 (Cell determinant and combining cells)**.** *For a cell J_j of length at most π/K, with loss mass m_j,*

$$\frac{\det \mathcal{A}_{J_j}}{T} \ge \frac{Q^2}{P^2} f(u_j), \qquad u_j = \frac{K m_j}{Q} \in [0, \pi]. \tag{25}$$

For any disjoint such cells in J,

$$\sqrt{c} \ge \frac{Q}{P} \sum_j g(u_j). \tag{26}$$

Proof. The comparison and the identity

$$\int_{x<y} q(x) q(y) \sin^2(K(y-x)) \, \mathrm{d}x \, \mathrm{d}y = \frac{1}{4} \left[m_j^2 - \left| \int q(x) e^{2iKx} \, \mathrm{d}x \right|^2 \right]$$

inserted in (20) prove (25). For positive semidefinite 2×2 matrices A, B, $\sqrt{\det(A+B)} \ge \sqrt{\det A} + \sqrt{\det B}$. For $A \succ 0$ this follows from $\operatorname{tr}(A^{-1/2} B A^{-1/2}) \ge 2\sqrt{\det(A^{-1/2} B A^{-1/2})}$; continuity covers singular A. Repeated application, including any additional absorption outside the selected cells, proves (26). □

Lemma 4.4 (Convexity and a global supporting tangent)**.** *The function g is convex on $[0, \pi/2]$. A tangent at any $0 < x \le 1/3$ lies below g on all of $[0, \pi]$.*

Proof. For $u > 0$,

$$g''(u) = \frac{H(u)}{f(u)^{3/2}}, \quad H(u) = 2\sin^2 u \, f(u) - (u - \sin u \cos u)^2, \quad H'(u) = 2 f(u) \sin(2u).$$

Since $H(0) = 0$, convexity holds on the stated half interval; also $g(0) = g'(0) = 0$ by $f(u) = u^4/3 + O(u^6)$. The identity

$$f(u) = 4 \int_0^u (u - v) \sin^2 v \, \mathrm{d}v$$

and the chord inequality $\sin v \ge 2v/\pi$ imply $f(u) \ge 4u^4/(3\pi^2)$ there. Moreover $f'(u) = 4 \int_0^u \sin^2 v \, \mathrm{d}v \le 4u^3/3$. Thus $g'(x) \le \pi x/\sqrt{3}$. Convexity gives the nonpositive tangent intercept $g(x) - x g'(x) \le 0$. For $v \in [\pi/2, \pi]$,

$$\frac{g(v)}{v} \ge \sqrt{1 - \frac{4}{\pi^2}} > \frac{\pi}{3\sqrt{3}} \ge g'(x).$$

The tangent is therefore below g on this second interval as well. In particular, using Jensen indiscriminately on all of $[0, \pi]$ would be incorrect; g is not convex on that entire interval. □

Theorem 4.5 (Finite resource floor)**.** *Let $U = KM/Q$, and for positive integers n set*

$$\beta_n = \frac{n^2 Q^2}{P^2} f(U/n).$$

Then, with $n_h = \max\{1, \lceil 2KL/\pi \rceil\}$,

$$\boxed{c \geq \beta_{n_h} \geq \frac{4k^4 M^4}{3Q^2(\pi + 2k\sqrt{P}\,L)^2} > 0.} \tag{27}$$

With $n_f = \max\{1, \lceil KL/\pi \rceil\}$, the stronger bound

$$\boxed{c \geq \beta_{n_f}} \tag{28}$$

holds if $U \leq \pi/2$ or $U/n_f \leq 1/3$. In particular, for fixed k, P, Q, M, (28) holds for every sufficiently large L, and uniformly over the admissible stacks

$$\liminf_{L\to\infty} (kL)^2 c \geq \frac{\pi^2 (kM)^4}{3PQ^2}. \tag{29}$$

Precisely, every sequence of admissible stacks with budgets $L_j \to \infty$ and the same k, P, Q, M obeys this lower limit with $L = L_j$.

Proof. Partition J into n_h equal cells. Every $u_j \leq \pi/2$, so (26) and ordinary Jensen give $c \geq \beta_{n_h}$. The lower estimate on f in Lemma 4.4 yields

$$\beta_{n_h} \geq \frac{4k^4 M^4}{3\pi^2 Q^2 n_h^2}.$$

Use $n_h \leq 1 + 2KL/\pi$ to obtain (27). For n_f, the cells have length at most π/K. If $U \leq \pi/2$, every u_j still lies in the convexity interval. Otherwise, when $U/n_f \leq 1/3$, sum the global supporting tangent at the common mean U/n_f. In either case $\sum g(u_j) \geq n_f g(U/n_f)$. Finally $n_f = KL/\pi + O(1)$ and $\beta_{n_f} = k^4 M^4/(3Q^2 n_f^2) + O(n_f^{-4})$, which proves (29). A stack shorter than L can be enclosed in an interval of length L by adding zero-loss vacuum, so the same uniform bound applies under a thickness cap. □

Remark 4.6 (Strictness and physical equality). For positive distributed loss these finite resource floors are not attained. Inside any positive-loss layer,

$$\mathrm{Im}(\bar{s}s')(z) = -k^2 \int_x^z q(v)|s(v,x)|^2 \,\mathrm{d}v$$

is nonzero after positive propagation. The forcing in the modulus equation is then positive, making (23) strict on a positive-measure set of loss pairs within a cell. At least one cell has such pairs. This already makes the resulting determinant bound strict. No characterization of a finite-L minimizing profile is asserted.

5 Sharp asymptotic thickness at exact fixed response

Fix a target in the strict active branch:

$$0 < T < 1, \qquad 0 < a < \frac{1 - T^2}{2}. \tag{30}$$

Define

$$b = 1 - T - a, \quad s = \sqrt{1-2a}, \quad y_0 = s - T, \quad x_0 = \sqrt{b^2 - y_0^2}, \quad D_0 = 2x_0, \quad \alpha = \frac{Ty_0}{sx_0}. \tag{31}$$

Here y_0, x_0, α are strictly positive, and D_0 is the established matrix optimum. Let $\mathcal{C}_L$ be the finite multilayers satisfying (18), total thickness at most L, transmitted power exactly T, and mean absorptance exactly a. Define

$$D_*(L) = \sup_{S \in \mathcal{C}_L} |A_L - A_R|$$

when the class is nonempty.

Theorem 5.1 (Sharp exact-target asymptotic)**.** *For fixed $k > 0$, $P > 1$, $Q, M > 0$, and targets (30), $\mathcal{C}_L$ is nonempty for every sufficiently large L, and*

$$\boxed{\lim_{L\to\infty} (kL)^2 [D_0 - D_*(L)] = \frac{\alpha \pi^2 (kM)^4}{3PQ^2}.} \tag{32}$$

Every finite admissible stack has $D < D_0$. The limit is sharp within the stated ordinary dielectric class, with exact fixed T, a, M.

This is the candidate contribution whose priority remains conditional. Its proof below is an ordinary mathematical proof, separately checked by two analytical reviewers. Numerical and symbolic checks in Section 8 are additional verification.

5.1 Uniform lower bound on the deficit

Write $x = |A_L - A_R|/2$ and

$$F(x) = a^2 - x^2 - 2T\left(b - \sqrt{b^2 - x^2}\right).$$

For every passive matrix, the phase optimization from Section 2 gives $Tc = \det \mathcal{A} \le F(x)$, and $x \le x_0 < b$. Moreover,

$$F(x_0) = 0, \quad F'(x) = -2x\left(1 + \frac{T}{\sqrt{b^2 - x^2}}\right), \quad F''(x) = -2 - \frac{2Tb^2}{(b^2 - x^2)^{3/2}} < 0.$$

The tangent inequality for the concave function F gives

$$F(x) \le [-F'(x_0)](x_0 - x) = \frac{sx_0}{y_0}(D_0 - 2x).$$

Thus the global, not merely local, inequality is

$$\boxed{D_0 - D \ge \alpha c.} \tag{33}$$

Combine this with the full-cell floor (29) to prove the lower limiting constant in (32), uniformly over $\mathcal{C}_L$.

There is also an explicit finite-thickness consequence. If β is any valid resource floor from Theorem 4.5, feasibility requires $a^2 \ge T\beta$, and

$$D \le 2\sqrt{b^2 - \left(\sqrt{1 - 2a + T\beta} - T\right)^2}. \tag{34}$$

This follows by solving $F(x) \ge T\beta$. It is an elementary determinant-constrained matrix envelope, not a separate new matrix theorem and not a claimed sharp finite-thickness Maxwell optimum.

5.2 A realizable sequence with the matching coefficient

Put $g_0 = kM$. For a positive integer m, use m cells of common physical length

$$d = \frac{\pi}{k\sqrt{P}}.$$

Each cell contains a centered lossy slice with $\varepsilon = P + iQ$, width $w = M/(Qm)$, and lossless real-P material on both sides. For sufficiently large m, $0 < w < d$. The integral of q is exactly M, and all material caps hold. This arrangement is a nodal-loss/Borrmann construction, an established physical principle [15, 16].

To compute it exactly, use dimensionless distance kz and the matrices

$$A = i\begin{pmatrix} 0 & 1 \\ P & 0 \end{pmatrix}, \quad B = -QE_{21}, \quad R = \exp\!\left(\frac{\pi A}{2\sqrt{P}}\right), \quad R^2 = -I.$$

Here E_{21} has its sole nonzero entry in row 2, column 1. With $\xi = kw = g_0/(Qm)$, define the algebraic factor

$$D(\xi) = e^{-A\xi/2} e^{(A+B)\xi} e^{-A\xi/2}. \tag{35}$$

The *physical* cell transfer is $C = RD(\xi)R$, because its spacers have positive dimensionless length $(\pi/\sqrt{P} - \xi)/2$. Append one ordinary real-P quarter-wave layer at each end. Then

$$RC^m R = (-1)^{m-1} D(\xi)^m.$$

The sign only changes transmission phase, so the stripped core may be taken as

$$F_m = D(g_0/(Qm))^m.$$

The negative exponents in (35) are an algebraic factorization; no negative-thickness layer is used.

The logarithm is analytic near $\xi = 0$. Since $D(-\xi) = D(\xi)^{-1}$, it is odd, and direct multiplication of the exponential series gives

$$\log D(\xi) = \xi B + \xi^3 C_3 + O(\xi^5), \qquad C_3 = \frac{1}{12}\begin{pmatrix} 0 & -Q \\ Q(P + 2iQ) & 0 \end{pmatrix}. \tag{36}$$

For completeness the cubic term is also $[A, [A, B]]/24 + [B, [A, B]]/12$, which evaluates to the displayed matrix. At fixed parameters,

$$m \log D(g_0/(Qm)) = G + m^{-2} H + O(m^{-4}), \quad G = -g_0 E_{21}, \quad H = \frac{g_0^3}{Q^3} C_3.$$

Since $G^2 = 0$, the Fréchet derivative of the exponential yields

$$F_m = F_0 + m^{-2} J + O(m^{-4}), \quad F_0 = I + G, \quad J = H + \tfrac{1}{2}(GH + HG) + \tfrac{1}{6} GHG. \tag{37}$$

In particular, $J_{12} = -g_0^3/(12Q^2)$. Now

$$\Sigma - F_0^\dagger \Sigma F_0 = \operatorname{diag}(2g_0, 0).$$

The order-m^{-2} term of the lower-right defect entry is $-2 \operatorname{Re} J_{12} = g_0^3/(6Q^2)$. Therefore

$$\boxed{c_m = \det(\Sigma - F_m^\dagger \Sigma F_m) = \frac{g_0^4}{3Q^2 m^2} + O(m^{-4}).} \tag{38}$$

This determines the invariant before matching the target powers.

5.3 Exact target matching with bounded-index dielectric layers

Let $p_0 = \sqrt{b - x_0}$, $r_0 = \sqrt{b + x_0}$, and $t_0 = \sqrt{T}$. The chosen real matrix optimizer is

$$S_0 = \begin{pmatrix} p_0 & t_0 \\ t_0 & -r_0 \end{pmatrix}.$$

The active-branch identities imply

$$r_0 - p_0 = 1 - s, \qquad T = (1 - r_0)(1 + p_0).$$

Its field transfer is triangular:

$$F_{\text{target}} = \begin{pmatrix} A_0 & 0 \\ -h_0 & A_0^{-1} \end{pmatrix}, \qquad A_0 = \frac{1 - r_0}{t_0} \in (0, 1), \quad h_0 = \frac{r_0 - p_0}{t_0} > 0.$$

With $\Lambda(s_j) = \operatorname{diag}(s_j, s_j^{-1})$, choose

$$s_{L,0}^2 = \frac{A_0 h_0}{g_0}, \qquad s_{R,0}^2 = \frac{A_0 g_0}{h_0}. \tag{39}$$

Then $\Lambda(s_{R,0}) F_0 \Lambda(s_{L,0}) = F_{\text{target}}$.

For a transformed limiting sheet, put

$$A = s_L s_R, \quad h = g_0 s_L / s_R, \quad v = A + A^{-1}.$$

Its response is exactly

$$T = \frac{4}{(v + h)^2}, \qquad a = \frac{2hv}{(v + h)^2}. \tag{40}$$

The target has $v - h = 2s/t_0 > 0$. The Jacobian determinant of the response with respect to v, h is

$$\det \frac{\partial(T, a)}{\partial(v, h)} = -\frac{16(v - h)}{(v + h)^5} \neq 0.$$

Also $dv/dA = 1 - A^{-2} \neq 0$ at $A_0 < 1$, while

$$\det \frac{\partial(A, h)}{\partial(\log s_L, \log s_R)} = -2Ah \neq 0.$$

The response map is therefore locally nonsingular.

To apply the implicit function theorem without treating the integer m as continuous, set $\tau = 1/m$ and analytically extend

$$\mathcal{F}(\tau) = \exp\left[\tau^{-1} \log D((g_0/Q)\tau)\right]$$

through $\tau = 0$. The expansion (36) makes the extension analytic, with $\mathcal{F}(0) = F_0$ and leading correction $O(\tau^2)$. The real implicit function theorem supplies positive real ratios

$$s_L(m) = s_{L,0} + O(m^{-2}), \qquad s_R(m) = s_{R,0} + O(m^{-2})$$

for which both actual powers T, a are *exactly* the targets for every sufficiently large integer m.

These diagonal transformations are physically implementable within (18). A quarter-wave layer of real refractive index n has transfer $\left(\begin{smallmatrix} 0 & i/n \\ in & 0 \end{smallmatrix}\right)$. Two consecutive such layers with indices n_1, n_2 give

$$-\operatorname{diag}(n_1/n_2, n_2/n_1).$$

For any fixed ratio $s > 0$, choose an integer j with $|\log s|/j < \frac{1}{2}\log P$, and set

$$n_1 = P^{1/4}e^{(\log s)/(2j)}, \qquad n_2 = P^{1/4}e^{-(\log s)/(2j)}.$$

Both indices lie strictly in $(1, \sqrt{P})$. Repeating this pair j times implements $\Lambda(s)$, up to a harmless sign. The strict inequalities persist under the small implicit-function adjustments. Thus a fixed finite number of passive, lossless dielectric layers realizes both matching networks smoothly, with total thickness $O(1/k)$ uniformly in m. No ideal transformer or singular surface is left in the physical stack. Coating invariance preserves (38) exactly.

5.4 Converting the invariant coefficient to the optimal deficit

After removing the harmless transmission sign, the matched scattering matrix converges to S_0 at order m^{-2}. All three limiting entries are nonzero. To avoid the branch discontinuity of a principal complex argument, define the unit complex number

$$\zeta = \frac{r_L r_R \bar{t}^2}{T|r_L r_R|}.$$

Then $\zeta + 1 = O(m^{-2})$, and the exact determinant identity is

$$\det \mathcal{A} = F(x) - 2T\sqrt{b^2 - x^2}(1 + \operatorname{Re}\zeta).$$

Since $|\zeta| = 1$, the phase penalty equals $T|r_L r_R||\zeta + 1|^2 = O(m^{-4})$. Taylor expansion about x_0, using $F'(x_0) = -2x_0 s/y_0 \neq 0$, and (38) therefore gives

$$D_0 - D_m = \alpha c_m + O(m^{-4}) = \frac{\alpha g_0^4}{3Q^2 m^2} + O(m^{-4}). \tag{41}$$

Including all spacers and matching layers, the physical thickness is

$$kL_m = \frac{\pi m}{\sqrt{P}} + O(1). \tag{42}$$

Choose a dimensionless constant C such that $kL_m - \pi m/\sqrt{P} \leq C$ for all sufficiently large m. For each large budget L, choose

$$m(L) = \left\lfloor \frac{\sqrt{P}(kL - C)}{\pi} \right\rfloor.$$

The corresponding stack fits within L, $m(L)/(kL) \to \sqrt{P}/\pi$, and it realizes the exact targets. Equation (41) proves the upper limiting constant in (32). Together with the uniform lower bound, this proves Theorem 5.1 for all large thickness budgets, not only a subsequence.

Corollary 5.2 (Asymptotic minimum thickness)**.** *Let $L_{\min}(\eta)$ be the infimum of thicknesses of admissible stacks with $D \geq D_0 - \eta$, keeping all material resources and target powers fixed. Then, as $\eta \downarrow 0$,*

$$kL_{\min}(\eta) \sim \frac{\pi(kM)^2}{Q}\sqrt{\frac{\alpha}{3P\eta}}. \tag{43}$$

Proof. The positive finite-thickness floor forces $L_{\min}(\eta) \to \infty$. The two matching asymptotic estimates for $D_0 - D_*(L)$, with arbitrarily small relative slack, can then be inverted. The constructed family supplies the upper estimate even if the supremum at a particular L is not attained. □

There is no finite-stack equality in (32)'s underlying active matrix ceiling; equality here means the optimal asymptotic coefficient of a supremum. The construction uses m lossy slices and $O(m)$ layers after merging adjacent identical spacers. It does not prove a minimum layer count, nor classify every asymptotically extremizing profile.

6 General channels, sharp extinction bounds, and a vector Maxwell counterexample

6.1 The incident field and its time reverse

At a fixed positive frequency, let the incoming and outgoing propagating channels be complete and power normalized. Incoming and outgoing bases are paired by physical time reversal. In these coordinates reciprocity is $S^T = S$, and passivity is $S^\dagger S \preceq I$. All radiation, polarization, and guided channels that can carry energy away must be included; an omitted radiation channel is not material absorption. Let S_0 be the scattering matrix of a specified lossless reciprocal reference background, so that $S_0^\dagger S_0 = I$ and $S_0^T = S_0$.

For a unit incoming vector u, the incident background solution has outgoing coefficients $S_0 u$. Reversing that entire background solution therefore gives the incoming vector

$$J_0 u = \overline{S_0 u} = S_0^\dagger \bar{u}. \tag{44}$$

The antiunitary J_0 is an involution. In general, it is not obtained by merely selecting an arbitrary second port. For plane waves in vacuum, $(\mathbf{E}, \mathbf{H})$ reverses to $(\mathbf{E}^*, -\mathbf{H}^*)$; propagation reverses and electric polarization is conjugated. For a lossy reciprocal scatterer, the resulting total scattered field is not the time reverse of the original total field.

Define dimensionless absorption, scattering, and extinction by

$$\begin{aligned} A(u) &= 1 - \|Su\|^2, \\ C(u) &= \|(S - S_0)u\|^2, \\ X(u) &= 2\,\mathrm{Re}\langle S_0 u, (S_0 - S)u\rangle = A(u) + C(u). \end{aligned} \tag{45}$$

Here the Hermitian inner product is conjugate linear in its first argument. These are powers divided by the incoming power, not the total outgoing power $\|Su\|^2$, which includes the unscattered incident wave.

Proposition 6.1 (Extinction reciprocity; established)**.** *Under the preceding assumptions,*

$$X(J_0 u) = X(u), \qquad A(u) - A(J_0 u) = -\big[C(u) - C(J_0 u)\big].$$

Proof. Writing $v = J_0 u$, one has $S_0 v = \bar{u}$, and transposing a scalar gives

$$\langle S_0 v, Sv\rangle = u^T S S_0^\dagger \bar{u} = u^\dagger (S_0^\dagger)^T S^T u = u^\dagger S_0^\dagger S u = \langle S_0 u, Su\rangle.$$

The penultimate equality uses both reciprocal symmetries. Substitution into (45) proves the result. The same proof applies to bounded operators on a Hilbert channel space when transpose is defined by the channel conjugation. It is the channel form of the established optical-theorem argument of Sounas and Alù [2, 1]. □

For finite-power wave packets or guided inputs, these normalizations apply directly. A plane wave on an infinite planar slab has infinite total power; there we use all powers per transverse area, divided by the incident flux per area. The specular plane-wave channels still obey (45). A cross section for a compact three-dimensional body, divided by incident *intensity*, has units of area and is a different normalization. The dimensionless bound below must not be applied to such a cross section without specifying an appropriate channel normalization or effective area.

6.2 A sharp bound at fixed common extinction

Theorem 6.2 (Channel extinction tradeoff; direct passivity consequence)**.** *For a reciprocal passive scattering operator, lossless reciprocal reference, and unit input u, let $v = J_0 u$ and $X = X(u) = X(v)$. Then*

$$0 \le X \le 4, \qquad |A(u) - A(v)| = |C(u) - C(v)| \le X - \frac{X^2}{4}. \tag{46}$$

For every $X \in [0, 4]$, this is attained by an abstract reciprocal passive three-port and an orthogonal reversed pair. For $0 < X < 4$, equality holds if and only if, after interchanging u, v if needed,

$$Su = \left(1 - \frac{X}{2}\right) S_0 u, \qquad \|Sv\| = 1. \tag{47}$$

At $X = 0$ both inputs satisfy $Su = S_0 u$; at $X = 4$ both satisfy $Su = -S_0 u$, with the corresponding assertion for v. Their absorptions then vanish.

Proof. Set $z = \langle S_0 u, Su \rangle$. Since $\|S_0 u\| = 1$ and S is a contraction, $|z| \le \|Su\| \le 1$. Hence $0 \le X = 2(1 - \operatorname{Re} z) \le 4$ and

$$0 \le A(u) = 1 - \|Su\|^2 \le 1 - |z|^2 \le 1 - (\operatorname{Re} z)^2 = X - X^2/4.$$

The same upper bound holds for $A(v)$, and two numbers in $[0, B]$ differ by at most B. Equality for $B > 0$ requires one number to equal B and the other to vanish. Equality in the two Cauchy–Schwarz steps is exactly (47); this proves necessity and sufficiency. The endpoint statements follow from $|z| \le \|Su\| \le 1$.

For sharpness set $t = 1 - X/2$, $s = \sqrt{1 - t^2}$, and

$$S = \begin{pmatrix} 0 & t & 0 \\ t & 0 & s \\ 0 & s & 0 \end{pmatrix}, \qquad S_0 = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}. \tag{48}$$

The defect is the positive semidefinite rank-one matrix

$$I - S^\dagger S = (s, 0, -t)^T (s, 0, -t),$$

so S is reciprocal and passive. For $u = e_1$, $v = e_2 = J_0 u$,

$$A(u) = 1 - t^2, \quad A(v) = 0, \quad X(u) = X(v) = 2(1 - t).$$

This attains (46). □

The proof uses only the familiar single-input absorption–scattering constraint and extinction symmetry. It is an *established consequence*, not a claim of a new reciprocity inequality. Its usefulness here is the exact choice of normalization and the sharp comparison with a physical class. A symmetric unitary dilation of (48) is

$$U = \begin{pmatrix} 0 & t & 0 & s \\ t & 0 & s & 0 \\ 0 & s & 0 & -t \\ s & 0 & -t & 0 \end{pmatrix}.$$

Terminating its fourth port in a matched absorber yields the three-port matrix. This is an ideal network realization; the following subsection gives a finite-coefficient Maxwell construction instead.

An additional elementary restriction concerns the illumination pair itself: for $\rho = |\langle u, J_0 u\rangle|$,

$$|A(u) - A(J_0 u)| \le \sqrt{1-\rho^2}. \tag{49}$$

Indeed $B = I - S^\dagger S$ satisfies $0 \preceq B \preceq I$. The rank-two Hermitian operator $uu^\dagger - vv^\dagger$ has its nonzero eigenvalues equal to $\pm\sqrt{1-|\langle u, v\rangle|^2}$. Taking its trace against B proves (49). This explains why a self-time-reversed incident field has zero asymmetry. No claim is made that the minimum of (49) and (46) is jointly sharp for every prescribed overlap.

6.3 The scalar two-port contrast constant

The full fixed-extinction envelope of the scalar problem is also elementary. For $X \in [0, 4]$, define

$$\tau = |1 - X/2|, \qquad q_X = \frac{1 + \sqrt{1 + 8\tau^2}}{4}.$$

Then the sharp scalar two-port value is

$$D_2(X) = 4\sqrt{q_X}(1 - q_X)^{3/2}. \tag{50}$$

To verify this, use the real contraction reduction in the proof below. For transmission magnitude σ, its attainable contrast is at most $4(1-q)\sqrt{q^2-\sigma^2}$, with $\sigma \le q \le 1$. Fixed X implies $\sigma \ge \tau$; increasing σ both reduces the integrand and shrinks the feasible interval. Therefore the best choice is the real transmission $t = 1 - X/2$. Maximizing over $q \in [\tau, 1]$ yields $2q^2 - q - \tau^2 = 0$, hence $q = q_X$. At that point $q^2 - \tau^2 = q(1-q)$, proving (50). Sharpness follows from

$$w = 1 - q_X, \quad d = \sqrt{q_X(1-q_X)}, \qquad S = \begin{pmatrix} w + d & 1 - X/2 \\ 1 - X/2 & w - d \end{pmatrix}.$$

Here $q_X \ge 1/2$ ensures $d \ge w$, and the eigenvalues are 1 and $1 - 2q_X$. The endpoint values $X = 0, 4$ follow by continuity or direct substitution. This envelope is another direct slice of the established two-port feasible set, and is symmetric under $X \mapsto 4 - X$.

Theorem 6.3 (Free-resource two-port contrast; established feasible-set consequence)**.** *For a reciprocal passive scalar two-port in vacuum-reference coordinates,*

$$S = \begin{pmatrix} r_L & t \\ t & r_R \end{pmatrix}, \qquad S_0 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix},$$

and for $X = 2(1 - \operatorname{Re} t) > 0$,

$$\frac{|A_L - A_R|}{X} \le \gamma_2 := \sqrt{10\sqrt{5} - 22} = 0.6005662120\ldots. \tag{51}$$

Let $q = (\sqrt{5}-1)/2$, $w = 1 - q = q^2$, and $d = q\sqrt{1-q^2}$. All equality matrices are, up to interchange of the two ports,

$$S = \begin{pmatrix} (w+d)e^{i\phi} & q^2 \\ q^2 & (w-d)e^{-i\phi} \end{pmatrix}, \qquad \phi \in \mathbb{R}. \tag{52}$$

Proof. Let $\tau = |t|$, $R = \max(|r_L|, |r_R|)$, and $r = \min(|r_L|, |r_R|)$. Passivity implies that the real symmetric matrix $\widetilde{S} = \left(\begin{smallmatrix} R & \tau \\ \tau & -r \end{smallmatrix}\right)$ is also a contraction. To see this, the diagonal entries of $I - \widetilde{S}^2$ are the original absorption diagonals, up to their order, while the magnitude of its off-diagonal entry is $\tau(R - r)$. The reverse triangle inequality bounds this by the original off-diagonal magnitude $|\bar{r}_L t + \bar{t} r_R|$. The two-by-two determinant criterion therefore preserves positive semidefiniteness.

Put $w = (R - r)/2$, $d = (R + r)/2$, and $q = \sqrt{d^2 + \tau^2}$. The eigenvalues of $\widetilde{S}$ are $w \pm q$, so $w + q \le 1$, and $|A_L - A_R| = 4wd$. Cases with zero contrast are immediate. Otherwise $0 < q < 1$, $\tau < 1$, and, writing $c = \tau/q$,

$$\frac{|A_L - A_R|}{X} \le \frac{2wd}{1-\tau} \le \frac{2q(1-q)\sqrt{1-c^2}}{1-qc} \le 2q\sqrt{\frac{1-q}{1+q}}.$$

The last inequality follows by squaring, since

$$(1-qc)^2 - (1-c^2)(1-q^2) = (c-q)^2 \ge 0.$$

The square of the final expression is $h(q) = 4q^2(1-q)/(1+q)$, and

$$h'(q) = \frac{8q(1-q-q^2)}{(1+q)^2}.$$

It has a unique maximum at $q = (\sqrt{5}-1)/2$, with $h(q) = 10\sqrt{5}-22$. Equality requires successively $t = \tau > 0$, $w = 1 - q$, $c = q$, and this value of q. At these magnitudes the positive-semidefinite defect has zero determinant. Thus the original reflection phases must attain reverse-triangle equality, namely $r_L r_R$ is negative real. This gives exactly (52). Direct substitution establishes sufficiency. □

This is a further optimization of the established two-port feasible region [4]; it should not be presented as an independent new fundamental law. Its strict difference from the multichannel supremum 1 is significant: permitting polarization conversion changes the resource-free answer even before introducing a more complicated spatial geometry.

There is also a phase issue when using dielectric stacks to approach this constant. Let $\hat{t}$ be transmission between the two physical boundary planes of a stack of thickness L. The coefficient in Theorem 6.3 is $t = e^{-ikL}\hat{t}$. Appending a vacuum spacer cannot change this reference-normalized phase. It can instead be adjusted with bounded thickness as follows. For any index cap $P > 1$, choose a positive integer m satisfying $m(1 - 1/\sqrt{P}) \ge 2$. An isotropic lossless slab with

$$1 \le n \le \sqrt{P}, \qquad d = \frac{m\pi}{kn}$$

is reflectionless at the working frequency, has port transmission $(-1)^m$, and multiplies normalized transmission by

$$e^{im\pi(1-1/n)}.$$

As n varies, this phase covers a full circle; $d \le m\pi/k$. The slab adds no absorption and preserves absorbed, reflected, and transmitted powers. It can change reference-relative scattered power and extinction; that phase-sensitive change is its purpose here. Consequently the ordinary mathematical dielectric construction of Theorem 5.1, applied to the interior target

$$T = q^4, \qquad a = 4q - 2,$$

and followed by this phase adjustment, approaches (51) with the same fixed material caps and loss budget. The target is in the strict active branch: $a > 0$ and $1 - T^2 - 2a = q^7 > 0$. This physical sharpness statement depends on the full physical hypotheses and proof of Theorem 5.1, not merely on any declaration with a similar name in Lean.

6.4 An exact two-layer vector Maxwell construction

Theorem 6.4 (Finite-coefficient planar realization and limiting sharpness)**.** *Fix $k > 0$. For every $-1 < t < 1$, $\eta > 0$, and $h > 0$, there is a normal-incidence reciprocal nonmagnetic planar structure comprising two homogeneous anisotropic dielectric layers, with identical vacuum exteriors, such that opposite y-polarized plane waves satisfy*

$$A_L = 0, \qquad A_R = (1-t^2)A_x(\eta, kh) > 0, \qquad X_L = X_R = 2(1-t). \tag{53}$$

The real permittivity eigenvalues lie in $[1, 4]$; the imaginary permittivity is positive semidefinite, with its nonzero eigenvalue equal to η in the first layer. Every layer has finite thickness and finite coefficients. Moreover, choosing $\eta_m = 1/m$ and $kh_m = m^2$ gives $A_x(\eta_m, kh_m) \to 1$. Thus for every prescribed $0 < X < 4$, the supremum of $|A_L - A_R|$ over this physical class is exactly $X - X^2/4$. The supremum of $|A_L - A_R|/X$, with $X > 0$ free, is 1.

Proof. Use the time convention $e^{-i\omega t}$, relative permeability $\mu = I$, and transverse coordinates (x, y). The first, absorbing layer has thickness h and relative permittivity

$$\epsilon_A = \mathrm{diag}(1 + i\eta, 1, 1).$$

Its two transverse polarizations decouple. The y polarization sees exact vacuum, whereas the x polarization sees a scalar slab of index $n = \sqrt{1 + i\eta}$, with $\mathrm{Re}\, n > 0$ and $\mathrm{Im}\, n > 0$.
For $s = \sqrt{1 - t^2}$, set

$$V = \begin{pmatrix} -t & s \\ s & t \end{pmatrix}, \qquad \epsilon_{W,\perp} = \frac{5I_2 - 3V}{2}, \qquad \epsilon_{W,zz} = 1,$$

with zero transverse–longitudinal entries. The second, lossless layer has thickness $d = \pi/k$. Since $V^T = V$ and $V^2 = I_2$, its transverse permittivity eigenvalues are 4 and 1, with eigenindices 2 and 1. Their optical phases are 2π and π. A finite isotropic slab between identical vacuum media has zero reflection at each integer multiple of π, independently of its impedance. Diagonalizing this anisotropic slab therefore gives zero reflection for every transverse polarization and port transmission

$$H = -V.$$

Its vacuum-reference transmission is $e^{-ikd}H = V$. This is an exact Fabry–Pérot half-wave plate, not a zero-reflection approximation.

For completeness, the scalar absorbing-slab coefficients are

$$r_0 = \frac{1-n}{1+n}, \quad p = e^{inkh}, \quad r_A = \frac{r_0(1-p^2)}{1-r_0^2p^2}, \quad \tau_A = \frac{(1-r_0^2)p}{1-r_0^2p^2}, \quad A_x = 1 - |r_A|^2 - |\tau_A|^2. \tag{54}$$

These follow by continuity of E and E' at the two interfaces. Since $\mathrm{Re}\, n > 0$, one has $|r_0| < 1$; also $|p| < 1$. The denominator in (54) is therefore nonzero, and $\tau_A \neq 0$. The exact flux identity is

$$A_x = k\eta \int_0^h |E_x(z)|^2 \, dz.$$

It implies $A_x > 0$: if the integral were zero, continuity would give $E_x = 0$ throughout the layer, and interface continuity would require both $1 + r_A = 0$ and $1 - r_A = 0$ for a unit incident wave, a contradiction. Also $A_x < 1$, since $\tau_A \neq 0$.

There are four accessible channels, ordered as (Lx, Ly, Rx, Ry). Put

$$R = \mathrm{diag}(r_A, 0), \qquad D = \mathrm{diag}(\tau_A, e^{ikh}).$$

Because the wave plate is exactly reflectionless, composing the two layers produces the exact four-port scattering matrix

$$S = \begin{pmatrix} R & DH \\ HD & HRH \end{pmatrix}, \qquad S_0 = e^{ik(h+d)} \begin{pmatrix} 0 & I_2 \\ I_2 & 0 \end{pmatrix}. \tag{55}$$

The symmetry of this matrix is explicit, and passivity follows either from layerwise flux balance or from the Maxwell flux identity with $\Im\epsilon \succeq 0$.

A left incident y wave is unchanged in the absorbing layer, and the lossless plate only changes its polarization: $A_L = 0$. A right incident y wave first crosses the plate, which produces $He_y = -se_x - te_y$. Only its x component is absorbed, so $A_R = s^2 A_x$. The co-polarized boundary-plane transmission in both directions is $-te^{ikh}$. Relative to vacuum propagation through $h + d$, this is

$$e^{-ik(h+d)}(-te^{ikh}) = t.$$

Consequently both extinctions are exactly $2(1 - t)$, proving (53). This calculation retains the incident reference phase; treating the port transmission as the normalized transmission would be incorrect.

It remains to establish the limit without replacing the slab by an ideal polarizer. As $m \to \infty$,

$$n_m = \sqrt{1 + i/m} \to 1, \qquad r_{0,m} \to 0, \qquad m\Im n_m = \frac{1}{2\,\mathrm{Re}\, n_m} \longrightarrow \frac{1}{2}.$$

The middle identity follows from $2\,\mathrm{Re}\, n_m \Im n_m = 1/m$. Hence $|p_m| = \exp[-m^2 \Im n_m] \to 0$. In (54), the denominator tends to 1, while $r_{A,m} \to 0$ and $\tau_{A,m} \to 0$. Thus $A_{x,m} \to 1$. For fixed $t = 1 - X/2$, equation (53) now gives

$$|A_L - A_R| \longrightarrow 1 - t^2 = X - X^2/4,$$

which matches the universal upper bound. Finally take $t_m = 1 - 1/m$, with $m \geq 2$. Then

$$\frac{|A_L - A_R|}{X} = \frac{1 + t_m}{2} A_{x,m} \longrightarrow 1.$$

Every member of either sequence is a finite-thickness Maxwell system with finite, passive, reciprocal material coefficients. □

This example supplies a rigorous realization of the multichannel envelope with ordinary anisotropic electric permittivity and two finite layers. The basic use of a polarizer followed by a reciprocal polarization transformer is established polarization optics; no priority claim for that mechanism is made. The construction is a useful exact synthesis and a counterexample to overgeneralizing the scalar theorem. Its transverse extent is infinite; it does not establish the same sharpness for compact three-dimensional bodies.

The real-permittivity ceiling 4 is not essential. For any prescribed $P > 1$, choose a positive integer j such that $p_j = [2j/(2j - 1)]^2 \leq P$, and instead set

$$\epsilon_{W,\perp} = \frac{p_j + 1}{2} I_2 - \frac{p_j - 1}{2} V, \qquad d = \frac{(2j - 1)\pi}{k}.$$

Its eigenindices are $2j/(2j - 1)$ and 1, with phases $2j\pi$ and $(2j - 1)\pi$. It has exactly the same port transmission $-V$, zero reflection, and normalized transmission V. All preceding observables and limits remain valid with real permittivity eigenvalues in $[1, P]$. The increased plate thickness is a fixed cost depending on P, independent of the approximation index m.

The point values used here admit passive causal dispersion. At the working frequency ω_0, the absorber value $\epsilon_x = 1 + i\eta$ is obtained from

$$\epsilon_x(\omega) = 1 + \frac{F}{\omega_0^2 - \omega^2 - i\gamma\omega}, \qquad F = \eta\gamma\omega_0, \quad \gamma > 0.$$

For the wave plate, an ideal lossless Lorentz oscillator with a pole $\Omega > \omega_0$ and positive strength $F = 3(\Omega^2 - \omega_0^2)$ gives the eigenvalue 4 at ω_0; the other eigenvalue is vacuum. Thus the construction does not require assigning a constant complex permittivity over all frequencies. The material bounds and sharpness asserted here hold at the working frequency, not over a finite band or at the separate lossless oscillator pole.

6.5 Finite-power beams: retaining the same sharp supremum

Corollary 6.5 (Finite-power version of the planar construction)**.** *For every $0 < X < 4$, laterally infinite stacks of the preceding form admit unit-power incident beams and their background time reverses with exactly that common extinction and with absorption difference approaching $X - X^2/4$. Consequently their relative absorption contrast has supremum 1 when extinction is also free. This statement concerns finite incident power, while retaining the laterally infinite material geometry.*

Proof. We give the continuity argument explicitly, since a plane wave is not a unit vector in a finite-power channel space. A planar stack conserves transverse wavevector $\kappa = (\kappa_x, \kappa_y)$. For $|\kappa| < k$, put $k_z = \sqrt{k^2 - |\kappa|^2}$, $\kappa_\perp = (-\kappa_y, \kappa_x)$, and $B_\kappa = I_2 - \kappa\kappa^T/k^2$. Since our tensors have $\epsilon_{zz} = 1$, no transverse–longitudinal coupling, and $\mu = I$, Maxwell's equations for the state $(E_x, E_y, H_y, -H_x)^T$ reduce within each layer to

$$\frac{d}{dz}\begin{pmatrix} E_t \\ H_t \end{pmatrix} = ik \begin{pmatrix} 0 & B_\kappa \\ \epsilon_\perp - \kappa_\perp \kappa_\perp^T/k^2 & 0 \end{pmatrix} \begin{pmatrix} E_t \\ H_t \end{pmatrix}.$$

Each layer transfer is therefore a finite matrix exponential, continuous in κ and in the plate angle. Vacuum admits the positive admittance matrix $Y_\kappa = (k_z/k)B_\kappa^{-1}$. The bases

$$E_t = \sqrt{2}\, Y_\kappa^{-1/2} a, \qquad H_t = \pm\sqrt{2}\, Y_\kappa^{1/2} a$$

give power $\|a\|^2$ and depend smoothly on κ near zero. They use fixed laboratory axes; an azimuth-dependent TE/TM basis, which is singular at normal incidence, is unnecessary.

The transfer-to-scattering denominator is invertible for real $|\kappa| < k$. Indeed, a nonzero null vector would give a solution with no incoming waves and only outgoing waves. Flux balance says its outgoing power plus nonnegative dissipated power is zero, so both outgoing amplitudes vanish. The zero boundary state and uniqueness of the finite-dimensional linear ODE then force the whole solution to vanish. Thus the four-port scattering matrices $S(\kappa, \theta)$, their lossless references, and their power observables are continuous near normal incidence. They are contractions. Here $t = \cos 2\theta$, with $0 < \theta < \pi/2$.

Choose $f \in L^2(\mathbb{R}^2)$, supported in the unit disk, with $\int |f|^2 = 1$, and take the left incoming angular spectrum $f_\sigma(\kappa)e_y = \sigma^{-1} f(\kappa/\sigma)e_y$, with $0 < \sigma < k$. Its background time reverse has right incident spectrum

$$e^{-ik_z(h+d)}\overline{f_\sigma(-\kappa)}e_y.$$

The sign reversal of κ and the background phase are part of time reversal. Parseval's identity and transverse translation invariance express each power as the integral of its fiber power against $|f_\sigma|^2$. Changing variables $\kappa = \sigma p$ and using the continuity just proved shows that the beam observables converge to (53) as $\sigma \downarrow 0$. For each fixed η, h, this convergence is uniform over any

compact plate-angle interval, by uniform continuity on a compact neighborhood of $\kappa = 0$. No uniform estimate in growing h or vanishing η is used.

For exact target matching, write $X = 2(1 - \cos 2\theta_0)$, where $0 < \theta_0 < \pi/2$. Choose shrinking intervals $[\theta_0 - \delta_m, \theta_0 + \delta_m]$ inside that open interval. Their normal-incidence extinctions strictly bracket X, because the derivative is $4 \sin 2\theta > 0$. For each fixed slab $\eta_m = 1/m, kh_m = m^2$, choose $\sigma_m > 0$ small enough that the beam extinctions retain these endpoint inequalities and the three power observables are within $1/m$ of their normal values uniformly over the angle interval. Continuity and the intermediate value theorem give θ_m in the interval with beam extinction exactly X. Extinction reciprocity supplies the same value for the reversed beam. Since $\theta_m \to \theta_0$ and $A_{x,m} \to 1$, the absorption difference tends to $X - X^2/4$. A diagonal sequence over fixed targets $X_j \downarrow 0$, choosing the absolute approximation error $o(X_j)$, proves that the relative contrast has supremum 1. □

6.6 Resource accounting and what the counterexample rules out

For the sequence in Theorem 6.4, the resources per transverse area are

$$\lambda_{\min}(\operatorname{Re}\epsilon) \geq 1, \quad \lambda_{\max}(\operatorname{Re}\epsilon) \leq 4, \quad \|\Im\epsilon\| \leq \frac{1}{m}, \quad kL_m = m^2 + \pi, \quad k\int \operatorname{tr}(\Im\epsilon)\, dz = m. \tag{56}$$

For the arbitrary-cap variant, replace the upper real-permittivity bound 4 by P and kL_m by $m^2 + (2j-1)\pi$. Thus material magnitudes are uniformly bounded, while both thickness and integrated loss grow. Any fixed positive loss cap Q is respected for all sufficiently large m. The construction does not contradict a theorem that fixes the integrated loss and thickness simultaneously.

A different obstruction already occurs at *one fixed finite* η, h, t, without taking any limit. The absorption matrix restricted to the two reversed y-polarized inputs is

$$\begin{pmatrix} 0 & 0 \\ 0 & (1-t^2)A_x \end{pmatrix}.$$

The off-diagonal entries vanish because the full absorption matrix is positive semidefinite and its left y diagonal is zero. Its determinant is exactly zero despite a positive amount of dissipative material, positive absorption in one direction, and finite geometry and coefficients. Indeed for any prescribed $M > 0$, choose a finite $\eta > 0$ and $h = M/\eta$, giving $\int \operatorname{tr}(\Im\epsilon)dz = M$. Therefore a direct vector analogue of the scalar positive Gram-determinant floor is false under a mere upper bound on material loss and a positive total loss budget. The missing hypothesis is control of loss-free field subspaces and their coupling to the illumination, not just more careful thickness accounting.

For these same finite structures $A_L = 0 < A_R$. Consequently no finite universal absorption ratio A_R/A_L exists under the stated general semidefinite-loss assumptions. If a ratio is defined only when its denominator is positive, add arbitrarily small loss in the y permittivity of the first layer. Continuous dependence of the explicit slab coefficients makes $A_L \downarrow 0$ while A_R tends to the positive value above; hence the ratio is still unbounded among systems with both absorptions positive. Each perturbed system remains reciprocal and passive.

When common extinction is fixed, the bound $|A_L - A_R|/X \leq 1 - X/4$ is nontrivial and sharp in the multichannel planar class. The supremum 1 without a fixed extinction is approached above as $X \downarrow 0$; it must not be described as near-unit absolute absorption at simultaneously negligible extinction. At $t = 0$, by contrast, the absolute absorption difference approaches 1, with fixed extinction $X = 2$ and normalized difference approaching $1/2$.

Scope of the resolution. The dimensionless fixed-extinction problem is completely resolved for abstract reciprocal channels and is sharply realizable in the full-vector planar class just described. A uniform improvement over the trivial relative contrast 1, or a finite absorption-ratio bound, is ruled out for any class containing these systems. This does not establish compact-body three-dimensional cross-section bounds, a uniform result for strictly coercive loss with fixed resources, or a finite-bandwidth law. The scalar inverse-square result in Section 5 is not asserted to survive those changes.

Numerical validation, not a substitute for proof. The script `vector_counterexample.py` evaluates (54)–(55) with NumPy and SciPy. For moderate thicknesses it independently constructs the exact four-by-four Maxwell fundamental matrix as $\exp[ikd\left(\begin{smallmatrix} 0 & I \\ \epsilon_\perp & 0 \end{smallmatrix}\right)]$ and converts it to scattering coefficients. Eight comparisons give a maximum matrix-entry discrepancy below 10^{-15}. Large-thickness limits use the exact stable slab formulas to avoid the ill-conditioning of exponentially growing transfer solutions. At $m = 128, t = 1/2$, it finds $X = 1$, $A_L = 0$, $A_R = 0.7499971390$, against the sharp upper value 0.75. At $m = 128, t = 1-1/m$, the relative contrast is 0.9960899503. The proof above, rather than these floating-point calculations, establishes the inequalities, exact zeros, and limits.

7 A sharp joint bound with prescribed time-reversal overlap

The separate inequalities $\Delta \le X - X^2/4$ and $\Delta \le \sqrt{1-\rho^2}$, where $\rho = |\langle u, J_0 u\rangle|$, do not give the sharp joint bound when $0 < \rho < 1$. We solve the resulting abstract operator problem below. The proof is an explicit positive-matrix certificate and makes no Maxwell-realizability claim for its equality operators. Priority of this particular joint tradeoff remains unestablished; it must not be described as new on the strength of a limited literature search.

Theorem 7.1 (Prescribed-overlap channel tradeoff)**.** *Fix $0 < \rho < 1$, $0 \le X \le 4$, and put*

$$d_\rho = \sqrt{1-\rho^2}, \qquad r = |1 - X/2|.$$

Consider all reciprocal passive scattering operators on channel spaces of arbitrary dimension, all lossless reciprocal reference operators, and all unit inputs satisfying $|\langle u, J_0 u\rangle| = \rho$ and common extinction X. The maximum absorption difference is

$$\Delta_\rho(X) = \begin{cases} d_\rho, & 0 \le r \le \rho/2, \\ d_\rho(1-\lambda^2)\sqrt{1-c^2}, & \rho/2 < r < \rho, \\ 1 - r^2, & \rho \le r \le 1, \end{cases} \tag{57}$$

where, in the middle branch, c is the unique number in $(0, \rho)$ satisfying

$$r = \frac{\rho + c - 2\rho c^2}{2 - \rho c - c^2}, \qquad \lambda = \frac{c(\rho - c)}{2 - \rho c - c^2}. \tag{58}$$

Two channels suffice in the first and middle branches; three suffice in the last branch. The formulas agree continuously at both transitions.

7.1 Canonical coordinates and a valid compression

First take reference $S_0 = I$, so the reverse is complex conjugation. A unit input and its conjugate can, after a global phase and a real orthogonal basis change, be written

$$u = pe_1 + iqe_2, \qquad \bar{u} = pe_1 - iqe_2, \qquad p^2 = (1+\rho)/2, \quad q^2 = (1-\rho)/2.$$

Indeed a phase makes $u^T u = \rho$ real and nonnegative; writing $u = x + iy$ then gives $x \cdot y = 0$, $\|x\|^2 = p^2$, and $\|y\|^2 = q^2$. For a general lossless reciprocal reference, an orthonormal basis fixed by its antiunitary involution J_0 gives this same form. Equivalently, in finite dimensions, Takagi factorization changes S_0 to identity by a unitary congruence. These basis changes preserve reciprocity, powers, extinction, and overlap.

Apply the further unitary congruence with $P = \mathrm{diag}(1, i, 1, \ldots)$. In the new coordinates the reference on the input plane is $J = \mathrm{diag}(1, -1)$, and the two incident vectors are real:

$$a = (p, q)^T, \qquad b = (p, -q)^T = Ja.$$

Let $\widehat{S}$ denote the full scattering operator in these coordinates, and let $\iota : \mathbb{R}^2 \to \mathcal{H}$ be the coordinate inclusion. Define real symmetric two-by-two matrices

$$B = \mathrm{Re}(\iota^\dagger \widehat{S} \iota), \qquad Q = \mathrm{Re}(\iota^\dagger \widehat{S}^\dagger \widehat{S} \iota), \qquad W = \begin{pmatrix} (1+\rho)/2 & 0 \\ 0 & -(1-\rho)/2 \end{pmatrix}. \tag{59}$$

The transpose symmetry of $\widehat{S}$ makes B symmetric. For every real vector x,

$$\|Bx\|^2 \le \|\iota^\dagger \widehat{S} \iota x\|^2 \le \|\widehat{S} \iota x\|^2 = x^T Q x \le \|x\|^2.$$

The first inequality follows by discarding the imaginary part of the output. Consequently

$$B^2 \preceq Q \preceq I_2. \tag{60}$$

This is a necessary relaxation for every number of channels; no claim that every such pair admits a reciprocal completion is needed.

Symmetry cancels the off-diagonal term in the forward overlap:

$$1 - X/2 = \mathrm{Re}\langle Ja, \widehat{S} a \rangle = \mathrm{tr}(WB), \qquad |A(a) - A(b)| = 4pq|Q_{12}| = 2d_\rho |Q_{12}|.$$

Replacing $\widehat{S}$ by $-\widehat{S}$ changes the sign of the first quantity and leaves Q fixed. Congruence by $\mathrm{diag}(1, -1)$ interchanges a, b, preserves W, and changes the sign of Q_{12}. Thus it suffices to analyze

$$r = \mathrm{tr}(WB) \ge 0, \qquad h = Q_{12} \ge 0, \qquad \Delta = 2d_\rho h. \tag{61}$$

7.2 A sum-of-squares certificate for the middle branch

Fix $0 < c < \rho$, and abbreviate

$$s = \sqrt{1 - c^2}, \qquad D = 2 - \rho c - c^2, \qquad \lambda = \frac{c(\rho - c)}{D}.$$

Here $D > 0$ and $0 < \lambda < 1$. Let O be the real rotation by $\theta \in (0, \pi/4)$ with $\cos 2\theta = c$, $\sin 2\theta = s$, and set

$$B_0 = O \begin{pmatrix} 1 & 0 \\ 0 & \lambda \end{pmatrix} O^T, \qquad Q_0 = B_0^2, \qquad r_0 = \mathrm{tr}(WB_0), \quad h_0 = (Q_0)_{12}. \tag{62}$$

Direct calculation gives

$$r_0 = \frac{\rho + c - 2\rho c^2}{D}, \qquad h_0 = \frac{1 - \lambda^2}{2} s.$$

To prove optimality over the full relaxation (60), define $E = \left(\begin{smallmatrix} 0 & 1/2 \\ 1/2 & 0 \end{smallmatrix}\right)$, so $\mathrm{tr}(EQ) = Q_{12}$, and define Z in the eigenbasis of B_0 by

$$O^T Z O = \begin{pmatrix} (\rho + c)/4 & -s/[2(1+\lambda)] \\ -s/[2(1+\lambda)] & (\rho - c)/(4\lambda) \end{pmatrix}, \qquad \nu = -\frac{s}{(1+\lambda)c} < 0. \tag{63}$$

All ingredients of the certificate can be checked explicitly:

$$\begin{aligned} \det Z &= \frac{D(\rho - c)}{16c(1-c^2)} > 0, \qquad Z \succ 0, \\ B_0 Z + Z B_0 &= W, \\ H := Z - \nu E &= \delta O \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} O^T \succeq 0, \qquad \delta = \frac{\rho + c}{4} + \frac{s^2}{2(1+\lambda)c} > 0. \end{aligned} \tag{64}$$

For example, the lower-right entry of the last identity uses exactly

$$(\rho - c)(1 + \lambda)c = 2\lambda(1 - c^2),$$

which is equivalent to the definition of λ. The off-diagonal entry follows from $O^T E O$ having off-diagonal $c/2$.

For any feasible real symmetric B, Q, expand a nonnegative square:

$$\begin{aligned} 0 &\le \operatorname{tr}\big[(B - B_0)Z(B - B_0)\big] \\ &= \operatorname{tr}(ZB^2) - \operatorname{tr}(WB) + \operatorname{tr}(ZB_0^2). \end{aligned}$$

Since $B^2 \preceq Q \preceq I$, $Z \succeq 0$, and $H \succeq 0$,

$$\begin{aligned} \operatorname{tr}(WB) &\le \operatorname{tr}(ZQ) + \operatorname{tr}(ZB_0^2) \\ &= \operatorname{tr}(HQ) + \nu h + \operatorname{tr}(ZB_0^2) \\ &\le \operatorname{tr}(H) + \nu h + \operatorname{tr}(ZB_0^2) = r_0 + \nu(h - h_0). \end{aligned} \tag{65}$$

The final equality holds because $H(I - Q_0) = 0$ and $\operatorname{tr}(WB_0) = 2\operatorname{tr}(ZB_0^2)$. If $r = r_0$, the negativity of ν forces $h \le h_0$. The reciprocal contraction $\widehat{S} = B_0$ in two channels attains equality. Transforming back by P^{-1} gives a complex symmetric contraction with reference identity.

It remains to verify that the parameter covers the middle strip once. With $r_0(c) = (\rho + c - 2\rho c^2)/D$,

$$r_0(0) = \rho/2, \quad r_0(\rho) = \rho, \quad r_0'(c) = \frac{2 + \rho^2 - 6\rho c + (1 + 2\rho^2)c^2}{D^2} > 0.$$

The numerator decreases on $[0, \rho]$ and its value at ρ is $2(1 - \rho^2)^2 > 0$. Thus every $\rho/2 < r < \rho$ has a unique $c \in (0, \rho)$, proving the middle branch of (57). An explicit stable expression is

$$c = \frac{2(2r - \rho)}{1 + \rho r + \sqrt{(1 + \rho r)^2 - 4(2\rho - r)(2r - \rho)}}.$$

Middle-branch equality. For the fixed sign conventions $r > 0, h > 0$, equality in (65) implies $B = B_0$ because $Z \succ 0$, and $Q = B_0^2$ because $Q - B^2 \succeq 0$ has zero trace against Z. Writing the compressed scattering block as $B + iC$, with C real symmetric, and its columns into the auxiliary space as F, gives

$$Q - B^2 = C^2 + \operatorname{Re}(F^\dagger F).$$

Its zero trace forces $C = 0$ and $F = 0$. Reciprocity also removes the reverse coupling from auxiliary channels. Therefore every middle-branch equality operator, in these canonical coordinates and after the two stated sign changes, has the form

$$\widehat{S} = B_0 \oplus S_{\text{aux}},$$

where S_{aux} is an arbitrary reciprocal contraction. This identifies all equality cases rather than merely one example.

7.3 The two outer branches and their transition conditions

The effect bound $0 \preceq Q \preceq I$ implies $|Q_{12}| \le 1/2$, hence $\Delta \le d_\rho$. Equality requires

$$Q = \frac{1}{2}\begin{pmatrix} 1 & \pm 1 \\ \pm 1 & 1 \end{pmatrix} =: P_\pm.$$

This follows, for example, by evaluating Q in the orthonormal vectors $(e_1 \pm e_2)/\sqrt{2}$: their quadratic values lie in $[0,1]$, and their difference is $2Q_{12}$. Endpoint equality forces them to be the eigenvectors with eigenvalues 1 and 0.

If $B^2 \preceq P_\pm$ and B is symmetric, the kernel of $P_\pm$ lies in the kernel of B, so $B = \alpha P_\pm$, $|\alpha| \le 1$. Thus $|\operatorname{tr}(WB)| \le \rho/2$. Conversely, for every $0 \le r \le \rho/2$, the symmetric contraction

$$\widehat{S} = e^{i\phi} P_+, \qquad \cos\phi = 2r/\rho,$$

has this value of r and $\Delta = d_\rho$. This proves the first branch, including necessity of its transition condition. Its full equality condition is the compression $Q = P_+$ or P_-, together with the prescribed forward overlap; the displayed matrix is a two-channel witness.

The general extinction bound gives $\Delta \le 1 - r^2$. For $0 < r < 1$, equality requires one reversed input, say b, to satisfy $\widehat{S}b = ra$, while $\|\widehat{S}a\| = 1$. Saturation of contractivity on a implies $\widehat{S}^\dagger \widehat{S} a = a$. Therefore

$$\rho = \langle a, b\rangle = \langle \widehat{S}a, \widehat{S}b\rangle = r\langle \widehat{S}a, a\rangle,$$

so $\rho \le r$. This proves necessity of the last-branch transition.

For sufficiency, work in $\mathbb{R}^3$. Let w be a unit vector in the input plane orthogonal to a, chosen so that $b = \rho a + d_\rho w$. Let e_3 be orthogonal to that plane and set

$$\alpha = \frac{\sqrt{1-r^2}}{d_\rho}, \qquad \beta = \frac{\sqrt{r^2-\rho^2}}{d_\rho}, \qquad f = \alpha w + \beta e_3, \qquad g = \beta w - \alpha e_3.$$

For $\rho \le r \le 1$, these are well-defined and (a, g, f) is an orthonormal basis. Define the real symmetric contraction by its matrix in that basis:

$$\widehat{S} = \begin{pmatrix} \rho/r & \sqrt{r^2-\rho^2}/r & 0 \\ \sqrt{r^2-\rho^2}/r & -\rho/r & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

Its two nonzero eigenvalues are 1 and -1. Since

$$b = \rho a + \sqrt{r^2-\rho^2}\, g + \sqrt{1-r^2}\, f,$$

it satisfies $\widehat{S}b = ra$, $\|\widehat{S}a\| = 1$, and common forward overlap r. Hence $\Delta = 1 - r^2$, proving the third branch. At $r = 1$, the difference is zero, as required. At $r = \rho$, the input plane reduces to the rank-one projection aa^T, and at $r = \rho/2$, the middle branch reduces to P_+. These agree with both outer formulas. For negative signed overlap $1 - X/2$, use $-\widehat{S}$. This completes the proof of Theorem 7.1.

For completeness, $\rho = 0$ recovers the sharp parabola of Theorem 6.2, with orthogonal reversed inputs. For $\rho = 1$, the two inputs differ only by a scalar phase, and their absorptions agree. Every $X \in [0, 4]$ is still feasible, for example with $S = (1 - X/2)S_0$.

7.4 An exact counterexample to taking the minimum of separate bounds

Let $\rho = 4/5$, $d_\rho = 3/5$, and $r = 59/95$, so $X = 72/95$. The middle-branch parameter is $c = 2/5$, $\lambda = 2/19$. The exact optimal real matrix in the $J = \mathrm{diag}(1, -1)$ reference coordinates is

$$B_0 = \frac{1}{190}\begin{pmatrix} 139 & 17\sqrt{21} \\ 17\sqrt{21} & 71 \end{pmatrix}.$$

Its eigenvalues are 1 and 2/19, and

$$\Delta_{4/5}(72/95) = \frac{1071\sqrt{21}}{9025} = 0.5438159079\ldots,$$

whereas the minimum of the separate bounds is 3/5. A compact independently checkable upper certificate is

$$W = \mathrm{diag}(9/10, -1/10), \quad Z = \begin{pmatrix} 7/8 & -17\sqrt{21}/168 \\ -17\sqrt{21}/168 & 3/8 \end{pmatrix}, \quad \nu = -\frac{19\sqrt{21}}{42}.$$

Here $\det Z = 19/168 > 0$, $B_0 Z + Z B_0 = W$, and

$$Z - \nu E = \frac{1}{8}\begin{pmatrix} 7 & \sqrt{21} \\ \sqrt{21} & 3 \end{pmatrix} \succeq 0, \qquad (Z - \nu E)(I - B_0^2) = 0.$$

Equation (65) therefore certifies the stated strict improvement without any numerical optimizer.

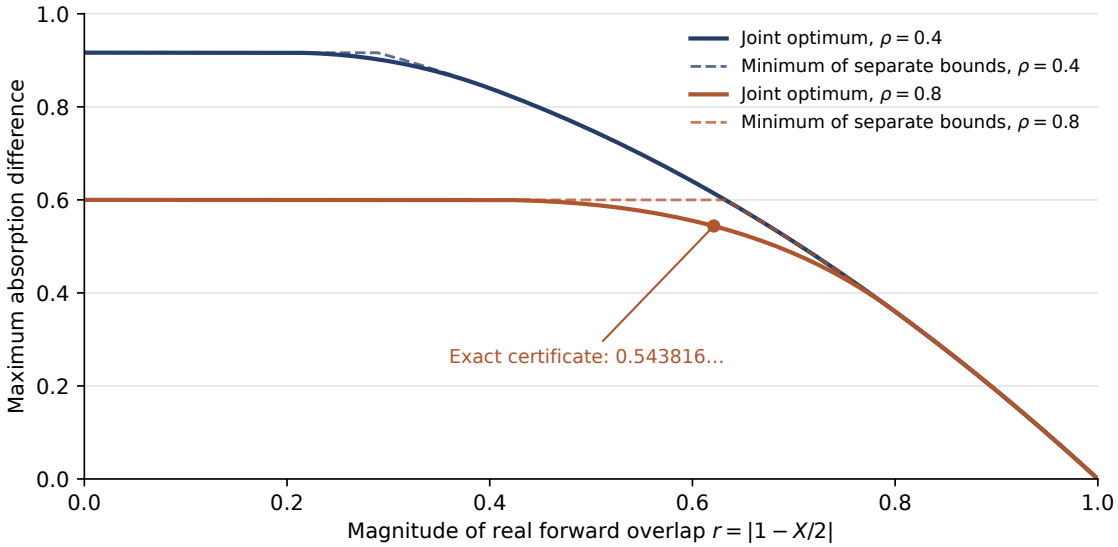


Figure 1: The joint theorem is stricter than taking the minimum of the two separate inequalities in the intermediate interval. Curves directly evaluate (57); they are not fits to optimization output. The marked point has the exact positive-matrix certificate displayed above. Both signs of $1 - X/2$ have the same bound.

Prior results and verification status. The plateau is the familiar pure-state trace-distance/Helstrom bound; the outer parabola is the single-input extinction inequality. The joint problem also resembles quantum information–disturbance tradeoffs. Buscemi and Sacchi [23], Eqs. (1)–(3) and (22), optimize average postmeasurement fidelity at prescribed discrimination success over quantum instruments. Their disturbance is not the coherent forward amplitude fixed here, and their optimization does not impose complex transpose symmetry on one

scattering contraction. D'Ariano et al. [24], Eqs. (12)–(13) and (26), instead optimize worst-case transformation probability and conditional-output fidelity. Their permitted output-unitary adjustments generally change our phase-sensitive forward overlap. These are exact differences in the statements, not proof that an equivalent reformulation is impossible.

Guo–Fan reflection feasibility [4] prescribes singular values and characterizes diagonal reflection coefficients. The joint problem here constrains $\mathrm{Re}\langle S_0 u, Su\rangle$ and the difference of two quadratic output powers at prescribed reversed-input overlap. The retrieved reflection theorem does not state (57); a reduction establishing equivalent prior coverage was not found. Our proof requires no assumed reciprocal completion, because its upper relaxation has explicit equality scattering matrices.

The complete ordinary proof, including all three branches and the equality conditions, survived an independent mathematical review. The general PSD trace certificate has also been verified in Lean, as specified in Appendix B. The canonical channel compression, parameter identities, and full theorem are not claimed Lean-verified. The priority classification is **CONDITIONAL**: this is a nontrivial proved joint refinement, but the retrieved comparisons do not establish a NEW result. No Maxwell equality realization at prescribed nonzero ρ is asserted.

8 Numerical and independent verification

8.1 A rational benchmark

Choose

$$P = 4, \quad Q = 2, \quad kM = 1, \quad T = \frac{12}{25}, \quad a = \frac{8}{25}.$$

Then

$$b = \frac{1}{5}, \quad s = \frac{3}{5}, \quad y_0 = \frac{3}{25}, \quad x_0 = \frac{4}{25}, \quad D_0 = \frac{8}{25}, \quad \alpha = \frac{3}{5},$$

and the limiting optimizer is

$$S_0 = \begin{pmatrix} 1/5 & 2\sqrt{3}/5 \\ 2\sqrt{3}/5 & -3/5 \end{pmatrix}, \qquad (A_L, A_R) = (12/25, 4/25).$$

Theorem 5.1 becomes

$$\boxed{\lim_{L\to\infty} (kL)^2 \left[\frac{8}{25} - D_*(L)\right] = \frac{\pi^2}{80}.} \tag{66}$$

For any admissible thickness, let $n = \max\{1, \lceil 2kL/\pi \rceil\}$. Here $U = 1 < \pi/2$, so the full-cell floor always applies:

$$c \geq \frac{1}{4}\left[1 - n^2 \sin^2(1/n)\right], \qquad \frac{8}{25} - D \geq \frac{3}{20}\left[1 - n^2 \sin^2(1/n)\right]. \tag{67}$$

The simpler universal consequence is

$$\frac{8}{25} - D \geq \frac{1}{5(\pi + 4kL)^2}.$$

For the construction, the core expansion is exactly

$$F_m = \begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix} + \frac{1}{m^2} \begin{pmatrix} 1/96 & -1/48 \\ (23+24i)/288 & 1/96 \end{pmatrix} + O(m^{-4}).$$

The matching ratios tend to $s_L = 1/\sqrt{3}, s_R = 1$. At these limiting ratios the response Jacobian with respect to $(\log s_L, \log s_R)$ is

$$\begin{pmatrix} 24/125 & 72/125 \\ 36/125 & -12/125 \end{pmatrix}, \qquad \det = -576/3125 \neq 0.$$

Their exact-target corrections satisfy

$$\log(s_L\sqrt{3}) = -\frac{25}{576m^2} + O(m^{-4}), \qquad \log s_R = \frac{13}{576m^2} + O(m^{-4}).$$

Consequently $c_m = 1/(12m^2) + O(m^{-4})$ and $D_0 - D_m = 1/(20m^2) + O(m^{-4})$.

Lossy slices m	$D_0 - D_m$	$m^2(D_0 - D_m)$	$m^2 c_m$	c_m/β_{full}
4	$3.13782636\,10^{-3}$	0.0502052218	0.0826755263	1.000405918
8	$7.82066448\,10^{-4}$	0.0500522527	0.0831687923	1.000106739
16	$1.95363761\,10^{-4}$	0.0500131229	0.0832921928	1.000027016
32	$4.88313325\,10^{-5}$	0.0500032845	0.0833230479	1.000006775
64	$1.22072318\,10^{-5}$	0.0500008214	0.0833307619	1.000001695
Limit		1/20	1/12	1

Table 2: Exact homogeneous-layer computations for the fixed-target family. The final column uses the interval from the first to the last lossy endpoint, with all lossless matching layers outside that interval, as permitted by Theorem 4.5. That interval is not the total stack thickness.

The calculation uses $k = 2\pi$, so lengths are in vacuum wavelengths. For $m = 4, 8, 16, 32, 64$, total thicknesses are approximately 2.643598, 3.644152, 5.644291, 9.644326, and 17.644335 wavelengths. These include all matching layers and an optional half-wave vacuum phase correction. The selected lossy spans are $(m-1)/4 + 1/(4\pi m)$; their $O(1)$ difference from total thickness does not change the limit (66).

Actual dielectric cascade powers differ from the specified T, a by less than 5×10^{-15}, and the physical cascade agrees with the matched field-transfer calculation within 3.5×10^{-15}. These residuals are floating-point checks, not a substitute for the implicit-function proof of exact target satisfaction.

8.2 Cross-checks, adversarial searches, and limiting cases

The Python/SciPy implementation uses stable scattering cascades, independent interface linear solves, characteristic transfer matrices, and analytic absorption integrals checked by numerical quadrature. In 250 random stacks with one to seven layers, the largest discrepancies were:

Check	Maximum discrepancy
Scattering cascade versus interface solve	8.4×10^{-16}
Characteristic matrix versus interface solve	7.9×10^{-16}
Absorption Gram integral versus $I - S^\dagger S$	1.3×10^{-15}
Analytic absorption integral versus quadrature	1.0×10^{-15}
Lossless unitarity	2.3×10^{-15}
Local determinant identity, relative error	8.8×10^{-14}

Separate SLSQP optimization of the abstract matrix envelope agreed within 8.0×10^{-11}, including both branches and zero-transmission and perfect-transmission limits.

The finite resource floor was challenged with 1200 random profiles, half embedded in additional passive surroundings. None violated the strengthened half-cell floor; the minimum ratio c/β_{n_h}

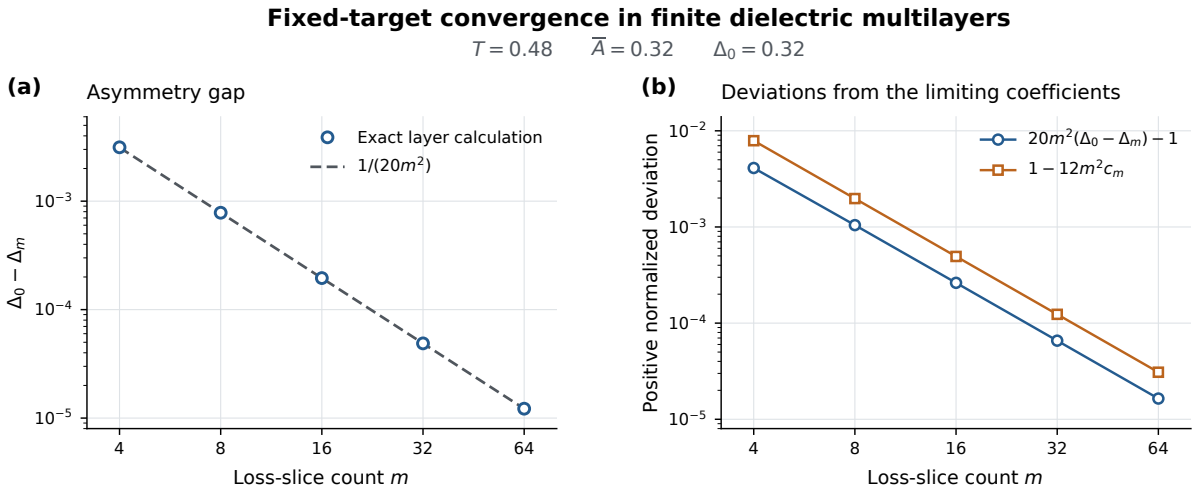


Figure 2: Fixed-target convergence from the existing numerical data. (a) The calculated asymmetry gap follows $1/(20m^2)$. (b) The positive normalized deviations $20m^2(D_0 - D_m) - 1$ and $1 - 12m^2c_m$ approach zero. Here m is the number of lossy slices, not thickness. The homogeneous-layer propagators were checked against an independent scattering cascade.

was 1.10173643. A separate set of 2000 complex-ODE comparisons at distances at most $\pi/(2K)$ gave minimum ratio 1.00001279. The extension of the comparison to the full distance π/K is justified by the proof, not inferred from that shorter-distance random sample.

For a uniform layer with $P = 4$, $L = 0.08$ vacuum wavelengths, and $M = QL$, the ratios c/β_1 at $Q = 1, 0.1, 0.01$ were respectively 1.0001588848, 1.0000015888, and 1.0000000159. Thus the short-cell bound has the correct weak-loss limiting coefficient. An eight-cell constrained SLSQP search at $P = 4, Q = 2, L = 0.1, M = 0.1$ found $c = 0.0123243316840$, above the floor 0.0123231683078. This nonconvex optimizer provides an adversarial check; it does not certify the finite-thickness optimum.

8.3 What was independently verified

One analytical agent solved and audited the matrix problem, then checked the full-cell refinement, exact matching, and all-L passage. A second independently reviewed the entire Maxwell argument and the general sharp constant. A separate numerical branch constructed actual dielectric stacks and checked them by independent exact layer formulations. The final LaTeX transcription received a further focused check for changed signs, constants, and assumptions.

The cubic logarithm and the general coefficient $g_0^4/(3Q^2)$ were also verified with exact SymPy polynomial arithmetic. The supplied script checks the general symbolic parameters, not only the rational benchmark. Analytic remainder estimates, uniform quantifiers, and the implicit function theorem remain parts of the written proof.

8.4 Reproduction and independent tests of the generalization

All four original JSON outputs reproduced byte for byte in the current environment. This preserves the original random-profile tests and limiting sequences. The separate generalization tests use independent implementations; the strongest cross-check solves a 12×12 modal interface-continuity system for all incident channels, with bounded internal propagation amplitudes. It does not import the implementation used to derive the closed formulas.

New independent check	Result
Normal-incidence vector systems	42
Largest full 4×4 scattering discrepancy	1.32×10^{-15}
Largest absorption/extinction discrepancy	8.89×10^{-16}
Largest interface-continuity residual	9.69×10^{-16}
Fixed-extinction complex optimizations	10 targets, 12 starts each
Largest discrepancy from scalar envelope	7.24×10^{-10}
Numerical normalized scalar maximum	0.6005662120015551
Exact-radical value, rounded	0.6005662120015561
Retuned finite-power beam checks	8

The optimizer's small discrepancies track its passivity-constraint tolerance; it is not a certified global optimizer. The beam quadrature retunes the plate angle and then checks the target at a finer quadrature order. Its small positive left absorption is consistent with convergence to the zero-absorption normal plane wave, not an assertion of exact zero for a finite-width beam. Complete numerical values are in `research/independent_checks.json`.

The overlap theorem received a separate exact symbolic check of sixteen parameter identities, including the positive certificate determinant and the monotone parameter derivative. Tests covered 42 middle-branch parameter pairs, 1,890 full complex symmetric contractions, and 42 outer equality matrices. A second implementation checked the exact rational example and 2,000 further complex contractions. The largest discrepancy from the independently optimized real equality family was 1.45×10^{-15}. The upper bound itself follows from the nonnegative-trace certificate, including a separately kernel-checked generic version, rather than from these numerical samples.

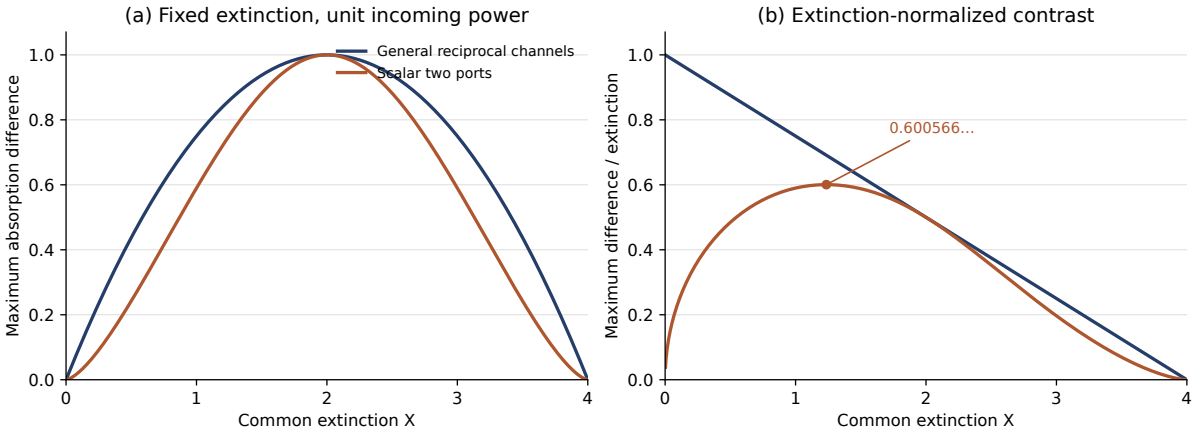


Figure 3: Direct evaluation of the proved formulas, without fitted curves. The general-channel envelope is sharp for reciprocal matrices and is approached by the vector planar Maxwell family. The scalar two-port envelope is smaller away from its common maximum. The normalized scalar contrast has the stated golden-ratio maximum; general reciprocal channels have supremum one as $X \downarrow 0$. In that latter limit the absolute absorption difference also tends to zero.

8.5 Recovered formal sources and the executed Lean audit

The available earlier source archive contains an explicitly unverified file with eleven real-algebra theorem statements. It contains neither a Lake project nor a declaration named `OQ7.theorem51`. Searches of the available original files and retained research record did not locate the separately reported project containing 692 public theorems. This is an availability finding: it does not show

that such a project cannot exist elsewhere. Its claims cannot be credited or audited without its source. All actually recovered originals are preserved, including the eleven-statement draft and the four numerical JSON files, which reproduced byte for byte.

A separate Lean supplement now provides an executed, kernel-checked baseline of 43 authored theorems. Its exhaustive census has 55 theorem constants, including twelve generated equations or auxiliary proofs. The precise statements and their boundaries are listed in Appendix B. In particular, the two-port matrix is an actual complex matrix, its absorption operator is actually $I - S^{\dagger}S$, and the finite-port theorem uses Mathlib's positive-semidefinite predicate. Extinction reciprocity and nonnegative absorbed power are derived from these definitions and hypotheses. The desired asymmetry inequality is not inserted as an assumption. The construction proving matrix sharpness also establishes complex positive semidefiniteness by an exact Gram factorization.

The verified scope does not include a Helmholtz solution, a dielectric admissibility predicate, a material-loss integral, the capped Fourier lemma, the uniform resource theorem, physical matching, or the limit of a thickness-constrained supremum. Consequently, neither Theorem 4.5 nor Theorem 5.1 is Lean-verified. The complete ordinary proofs and the analytic bridges in Appendix A specify the remaining work; they are not substitutes for it.

Toolchain and executed build. The project pins Lean 4.19.0 [21] and Mathlib commit `c44e0c8ee63ca166450922a373c7409c5d26b00b` [22]. All transitive revisions are recorded in `lean/lake-manifest.json`. The unmodified standalone Lean executable could not discover its installation path in the managed Linux environment; the official Lake executable did run. The source-visible default Lake target uses the supported `Lean.Elab.runFrontend` API with trust level zero, checks its success result, and writes genuine compiled modules. Separate sequential frontend processes keep memory bounded. No compiler or kernel was modified. The default `lake build` elaborates every supplied proof module and executes the axiom audit; it is not an empty build target.

After selecting the pinned toolchain, the clean commands are

```
lake clean oq7Audit
lake build
```

The qualified clean removes this project's compiled artifacts while retaining the pinned upstream dependency cache. The setup script in `lean/scripts/bootstrap-linux.sh` downloads and checksums the official compiler, resolves the pinned dependencies, obtains their upstream cache, and runs these commands on Linux or WSL2. The tested default target differs from invoking the standalone-compiler `lake build OQ7` target.

Axioms and failure controls. Every authored theorem has an explicit `#print axioms` command. An additional command enumerates all `OQ7` theorem constants from the elaborated environment, records their full types and transitive axiom dependencies, and rejects an empty census or an axiom outside `propext`, `Classical.choice`, and `Quot.sound`. These are the standard classical Lean/Mathlib foundations; no additional mathematical axiom, `sorryAx`, or native-decision proof axiom is accepted. The archived negative controls demonstrate that an invalid proof, an empty audit, and an admitted false theorem injected into a temporary copy of the full project each fail with a nonzero exit code. The admitted theorem is detected specifically through `sorryAx`. Invalid control sources are not part of the proof modules.

The clean log, per-module logs, negative controls, source hashes, and complete machine-readable theorem census are included in the reproducibility package. The exact declarations and observed logs take precedence over a prose description of their scope.

9 Obstructions, scope, and final verdict

9.1 Failed shortcuts and why they fail

A layer-by-layer bound loses the required information. For a homogeneous slab, write $\sqrt{p+iq} = \nu + i\kappa$, so $q = 2\nu\kappa$. Substituting its exact homogeneous-layer propagator into (21) gives

$$c_{\text{slab}} = \frac{4}{\nu^2+\kappa^2}\left[\nu^2 \sinh^2(k\kappa\ell) - \kappa^2 \sin^2(k\nu\ell)\right].$$

Expansion at small thickness ℓ gives

$$c_{\text{layer}} = \frac{q^2(k\ell)^4}{3} + O(\ell^6)$$

at fixed material coefficients. Dividing the same material into N artificial sublayers and using only $(\sum_j \sqrt{c_j})^2$ produces a bound proportional to N^{-2}, which collapses under refinement at fixed physical profile. The continuum pair identity retains the cross-location field information that this shortcut discards.

Unrestricted full-period Jensen is false. The function $g(u) = \sqrt{u^2 - \sin^2 u}$ is not convex throughout $[0, \pi]$. The small-mean global tangent in Lemma 4.4 is necessary for the full-period proof at arbitrary fixed total mass.

Loss amount alone gives no positive uniform gap. If the loss-density cap is removed, take a single slab $\varepsilon = P + iq$ of width M/q, and let $q \to \infty$. Its exact transfer approaches $\left(\begin{smallmatrix} 1 & 0 \\ -kM & 1 \end{smallmatrix}\right)$; hence $c \to 0$, while $T \to 4/(2+kM)^2 > 0$. More precisely $c \sim k^4M^4/(3q^2)$. If thickness is unrestricted while P, Q, M remain fixed, the family in Section 5 also has $c \to 0$ and reaches the desired matrix response asymptotically. If the real-permittivity cap is removed, a homogeneous slab with fixed q, ℓ and $p \to \infty$ has $c \sim q^2(k\ell)^2/p \to 0$; this last bare-slab example does not hold arbitrary T, a fixed. These distinguish essential resources from mere passivity.

Positive absorption does not exclude every matrix optimizer. In the high-mean branch, an optimizer with one zero reflection has $\det \mathcal{A} = 2a - 1 + T^2 > 0$ above the interface. The strict gap derived here concerns (30); positive definiteness alone does not exclude all high-mean optimizers.

9.2 What remains unresolved

The exact optimal envelope at every finite L, the complete set of asymptotically optimizing profiles, and a necessary minimum layer count remain open in this investigation. The strict limits $T = 0$, $a = 0$, and $a = (1 - T^2)/2$, as well as the material case $P = 1$, are excluded from Theorem 5.1; its Jacobian or envelope arguments can degenerate there. A bound valid with a fixed positive minimum feature size or a finite catalogue of dispersive materials would be a different realizability problem.

The present result is monochromatic. The hypotheses prescribe material values at one frequency and do not specify dispersion over a frequency interval, so they do not establish a nontrivial finite-bandwidth or causality tradeoff. The resource-free vector planar and finite-power channel formulations are addressed in Section 6. Sharp compact-body three-dimensional plane-wave cross-section bounds, controlled finite apertures, and quantitative coercive-loss bounds remain unresolved.

Verdict	Precise scope
KNOWN	Fixed-T, a two-port feasibility and optimum; the Rollett invariant; the channel passivity envelope; scalar contrast optimizations; polarization-filter mechanisms.
PROVED, ordinary	Exact finite-coefficient vector dielectric construction and its finite-power beam limit, with exact target extinction; counterexamples to an unrestricted finite absorption-ratio bound and a tensor-loss determinant floor. Physical statements are not Lean-verified.
CONDITIONAL priority	The restricted scalar Theorem 5.1 and prescribed-overlap Theorem 7.1, each with a complete independently reviewed ordinary proof. No equivalent theorem was located; priority is not certified, and complete Lean verification is absent.
LEAN-VERIFIED subset	Only the exact algebraic and finite-channel declarations and generic PSD certificate in Appendix B; the formal model and its hypotheses are part of each claim.
UNRESOLVED	Compact-body three-dimensional cross-section sharpness, finite-bandwidth tradeoffs, coercive tensor-loss resource bounds, exact finite-thickness scalar optima, Maxwell equality realization for prescribed nonzero illumination overlap, and the unavailable full Lean project.

The broad tutorial question is not declared closed. The present results resolve precise channel and planar formulations and identify the constraints missing from broader resource statements. The reproducibility package contains unchanged original artifacts, a separate source revision, full numerical records, independent audits, the pinned Lean project, and its actual build and axiom logs. It does not contain a proof of the missing declaration `OQ7.theorem51`.

A Detailed analytic bridges for the restricted theorem

This appendix records ordinary proofs of the steps most relevant to a future complete formalization of Theorem 5.1. None of the arguments in this appendix is represented as already verified in Lean. They preserve the original theorem and replace avoidable analytic complications by explicit identities. The earlier manuscript remains unchanged in the source archive.

A.1 A first-zero proof of the complex comparison

Fix the initial point at zero, and write $s = s(h, 0)$, $\rho = |s|$, and $j = \mathrm{Im}(\bar{s}s')$. Since $s(0) = 0$ and $s'(0) = 1$, one has $s(h)/h \to 1$ and $s'(h) \to 1$ as $h \downarrow 0$. In particular, $\rho > 0$ initially and

$$\rho(h) \to 0, \qquad \rho'(h) = \frac{\mathrm{Re}(\bar{s}(h)s'(h))}{|s(h)|} \to 1.$$

On any interval before its first positive zero, differentiation of $\rho^2 = \bar{s}s$ and the Helmholtz equation give

$$\rho'' + K^2\rho = (K^2 - k^2 p)\rho + \frac{j^2}{\rho^3} \geq 0.$$

These equations hold inside every homogeneous layer; ρ, ρ' are continuous at interfaces as long as $\rho > 0$.

Suppose a first zero h_0 existed in $(0, \pi/K)$. For $0 < \delta < h < h_0$, variation of constants, with its nonnegative sine kernel, yields

$$\rho(h) \geq \rho(\delta)\cos K(h - \delta) + \frac{\rho'(\delta)}{K}\sin K(h - \delta).$$

Only bounded, piecewise continuous coefficients on $[\delta, h]$ are used here; there is no division by the initial zero. Letting $\delta \downarrow 0$ gives $\rho(h) \geq \sin(Kh)/K$. Then $h \uparrow h_0$ contradicts $\rho(h_0) = 0$. There is consequently no such first zero, and the same argument proves the comparison throughout $(0, \pi/K)$. The two endpoints follow by continuity. This proof avoids both a Poincaré theorem for complex fields and an estimate of the singular-looking forcing at the initial point.

A.2 A positive-part proof of the capped Fourier bound

Extend q by zero to a full period of length π/K, let $\mu = \int q$, and put $\lambda = K\mu/Q \in [0, \pi]$. For any phase ϕ and $c = \cos\lambda$, the pointwise inequality

$$q(z)\cos(2Kz - \phi) \leq cq(z) + Q\max\{\cos(2Kz - \phi) - c, 0\}$$

follows separately on the sets where the expression in braces is positive and nonpositive. The positive arc has half-width λ; its integral over one period is

$$\int \max\{\cos(2Kz - \phi) - \cos\lambda, 0\}\,\mathrm{d}z = \frac{\sin\lambda - \lambda\cos\lambda}{K}.$$

Integration cancels the mass term because $\mu = Q\lambda/K$, giving

$$\mathrm{Re}\left(e^{-i\phi}\int q(z)e^{2iKz}\,\mathrm{d}z\right) \leq \frac{Q}{K}\sin\lambda.$$

Choose ϕ to align a nonzero moment with the positive real axis; the zero moment is immediate. This proves Lemma 4.2, including the zero-mass and full-period endpoints, without an existence theorem for an optimizing measurable density.

A.3 Finite products with an explicit remainder

All norms in this subsection are submultiplicative operator norms, with $\|I\| = 1$. Fix $P, Q, g_0 > 0$. Let $A, B, D(\xi), C_3$ be as in Section 5. Taylor's theorem for the three matrix exponentials, followed by finite multiplication, gives

$$D(\xi) = I + \xi B + \xi^3 C_3 + \frac{\xi^4}{2}(BC_3 + C_3B) + O(\xi^5). \tag{68}$$

The quadratic coefficient vanishes because $B^2 = 0$. The cubic coefficient is the displayed matrix in (36); it can equally be obtained from

$$[A,[A,B]] = \begin{pmatrix} 0 & -2Q \\ 2PQ & 0 \end{pmatrix}, \qquad [B,[A,B]] = \begin{pmatrix} 0 & 0 \\ 2iQ^2 & 0 \end{pmatrix}.$$

For the quartic coefficient, write the first four unknown coefficients and compare powers in the exact identity $D(-\xi)D(\xi) = I$. The order-four equation is precisely twice that coefficient minus $BC_3 + C_3B$. Matrix exponential series converge absolutely, so the remainder in (68) has a norm bound $C|\xi|^5$ on a sufficiently small fixed interval.

Set

$$G = -g_0E_{21}, \quad H = \frac{g_0^3}{Q^3}C_3, \quad K_4 = \frac{GH + HG}{2}, \quad A_m = I + \frac{G}{m}, \quad D_m = D\left(\frac{g_0}{Qm}\right).$$

Then $G^2 = 0$, and

$$D_m = A_m + W_m, \qquad W_m = \frac{H}{m^3} + \frac{K_4}{m^4} + E_m, \quad \|E_m\| \leq \frac{C_E}{m^5}.$$

There are fixed constants C_0, C_W such that $\|A_m\|, \|D_m\| \le 1 + C_0/m$ and $\|W_m\| \le C_W/m^3$ for all sufficiently large m. Telescoping twice gives the exact noncommutative identity

$$D_m^m - A_m^m - \sum_{j=0}^{m-1} A_m^{m-1-j} W_m A_m^j = \sum_{j=0}^{m-2} \sum_{\ell=0}^{m-2-j} D_m^{m-2-j-\ell} W_m A_m^\ell W_m A_m^j.$$

There are $m(m-1)/2$ terms on the right. Its norm is therefore at most

$$\frac{C_W^2 e^{C_0}}{2m^4}.$$

For an arbitrary matrix Y, nilpotence gives the exact single-insertion sum

$$\sum_{j=0}^{m-1} A_m^{m-1-j} Y A_m^j = mY + \frac{m-1}{2}(GY + YG) + \frac{(m-1)(m-2)}{6m} GYG. \tag{69}$$

Since $GK_4 + K_4G = GHG$ and $GK_4G = 0$, substitution of $H/m^3 + K_4/m^4$ into (69) gives exactly

$$\frac{1}{m^2}\left[H + \frac{GH + HG}{2} + \frac{GHG}{6}\right] - \frac{GHG}{6m^4}.$$

The contribution from E_m has norm at most $C_E e^{C_0}/m^4$. Also $A_m^m = I + G$. Thus

$$\left\| D_m^m - I - G - \frac{J}{m^2} \right\| \le \frac{\|GHG\|/6 + C_E e^{C_0} + C_W^2 e^{C_0}/2}{m^4}, \qquad J = H + \frac{GH + HG}{2} + \frac{GHG}{6}.$$

This proves (37) with its required uniform remainder at fixed parameters. The isolated term $-GHG/(6m^4)$ is not claimed to be the complete quartic coefficient: multiple insertions also contribute at that order. In particular, no matrix logarithm is needed to justify the invariant coefficient in (38).

A.4 Exact matching as a real finite-dimensional implicit problem

The implicit function argument can treat the eight real coordinates of the complex matrix F as independent parameters. It need not interpolate the integer number of layers. Let

$$W = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \qquad K = W^{-1} \Lambda(e^{\ell_R}) F \Lambda(e^{\ell_L}) W.$$

On the open set $K_{22} \neq 0$, define smooth scattering expressions

$$r_L = -K_{21}/K_{22}, \qquad r_R = K_{12}/K_{22}, \qquad t_R = 1/K_{22}, \qquad t_L = \det K / K_{22}.$$

These are the usual two-sided boundary formulas. Define the real response extension

$$\widetilde{T} = |t_L|^2, \qquad \widetilde{a} = 1 - \frac{|r_L|^2 + |r_R|^2 + |t_L|^2 + |t_R|^2}{2}.$$

Off the determinant-one locus it is not asserted that this extension describes a reciprocal physical medium. On the actual transfer sequence, $\det F_m = 1$, the two transmission amplitudes agree and these expressions are exactly the physical T, a.

At $F_0 = I - g_0 E_{21}$, put $A = e^{\ell_L+\ell_R}$, $h = g_0 e^{\ell_L-\ell_R}$, and $v = A + A^{-1}$. The determinant of the response derivative with respect to (ℓ_L, ℓ_R) factors as

$$\det \frac{\partial(T, a)}{\partial(\ell_L, \ell_R)} = \left[-\frac{16(v-h)}{(v+h)^5}\right](1 - A^{-2})(-2Ah).$$

At the strict target, $A = A_0 \in (0, 1)$, $h = h_0 > 0$, and $v-h = 2s/\sqrt{T} > 0$, so all factors are nonzero. The real parameter implicit function theorem gives a differentiable map $F \mapsto (\ell_L(F), \ell_R(F))$ near F_0 for which the two extended responses equal the specified targets. Because $F_m - F_0 = O(m^{-2})$, differentiability yields

$$\ell_L(F_m) - \ell_L(F_0) = O(m^{-2}), \qquad \ell_R(F_m) - \ell_R(F_0) = O(m^{-2}).$$

Evaluating this map on the actual unimodular sequence supplies exact physical target matching for every sufficiently large integer m. The fixed quarter-wave realizations in Section 5 then realize these ratios with bounded additional thickness.

A.5 Phase control and the passage to every large budget

The original principal-phase expression requires a locally continuous choice of argument near a negative real number. The revised proof instead uses

$$\zeta = \frac{r_L r_R \bar{t}^2}{T|r_L r_R|}, \qquad |\zeta| = 1, \qquad \zeta + 1 = O(m^{-2}).$$

Thus $1 + \operatorname{Re}\zeta = |\zeta + 1|^2/2$, without any branch choice. This is the sole phase repair; no target or asymptotic constant changes.

Finally, let C bound the added dimensionless thickness uniformly. For every sufficiently large real budget L, the integer $m(L) = \lfloor \sqrt{P}(kL - C)/\pi \rfloor$ is admissible and tends to infinity with $m(L)/(kL) \to \sqrt{P}/\pi$. The constructed asymmetry is a lower bound on $D_*(L)$; the cell inequality bounds every admissible asymmetry from above. The supremum exists as a finite real number once nonemptiness is established, because $0 \le D \le 1$. These two inequalities prove the limit for all large budgets, irrespective of whether a finite-budget optimizer exists.

B Manuscript–Lean correspondence and reproducibility

All declaration names below have the prefix `OQ7.`. The completed supplement contains 43 explicitly authored theorems: eleven in `Algebra`, eight in `Scattering`, five in `Universal`, seven in `FinitePorts`, eight in `Sharpness`, and four in `OverlapCertificate`. The environment also contains eight generated definition equations and four auxiliary proof constants, for fifty-five theorem constants in this namespace. Generated constants are audited but are not counted as independent results. The complete names, types, and dependencies are in `lean/logs/public_theorems.json`.

B.1 Exact finite-port theorem represented in Lean

For a finite index type n, $S \in \mathbb{C}^{n\times n}$, a map $\rho : n \to n$, and two indices j, k, the definitions are

$$A_j = 1 - \sum_i |S_{ij}|^2, \quad C_j = \sum_i |S_{ij} - \mathbf{1}_{i=\rho(j)}|^2, \quad X_j = A_j + C_j.$$

The declaration `OQ7.finite_port_absorption_asymmetry` assumes exactly

$$S^{\mathsf{T}} = S, \qquad I - S^{\dagger}S \succeq 0, \qquad \rho(j) = k, \quad \rho(k) = j,$$

and concludes $|A_j - A_k| \le X_j - X_j^2/4$. Thus the tested reference columns are unit port vectors interchanged by ρ. The theorem does not quantify over arbitrary coherent inputs or an arbitrary symmetric unitary reference S_0. Those broader time-reversal and Maxwell identifications are part of the ordinary proof of Proposition 6.1, not this formal declaration.

The declaration `OQ7.finite_port_bound_attained` quantifies over every real $X \in [0, 4]$ and constructs a complex 3×3 symmetric matrix with $I - S^\dagger S \succeq 0$, the specified extinction, and absorption difference $X - X^2/4$. Its fixed reference map swaps ports zero and one and fixes port two. It is a complete sharpness theorem for that matrix class. Neither a dielectric realization nor the equality classification for all Hilbert-space operators is part of its statement.

B.2 Exact PSD certificate represented in Lean

For finite real matrices, let B, B_0 be symmetric, put $W = B_0 Z + Z B_0$ and $Y = Z - \nu E$, and assume

$$Z \succeq 0, \quad Q - B^2 \succeq 0, \quad I - Q \succeq 0, \quad Y \succeq 0, \quad Y B_0^2 = Y.$$

There is no sign restriction on ν in the formal certificate. The declaration `OQ7.overlap_certificate_identity` proves the exact identity

$$\begin{aligned} &\operatorname{tr}(W B_0) + \nu\{\operatorname{tr}(EQ) - \operatorname{tr}(EB_0^2)\} - \operatorname{tr}(WB) \\ &\quad = \operatorname{tr}[Z(Q - B^2)] + \operatorname{tr}[Y(I - Q)] + \operatorname{tr}[(B - B_0) Z (B - B_0)]. \end{aligned}$$

The declaration `OQ7.overlap_psd_trace_certificate` proves that the left side is nonnegative from the stated matrix hypotheses. Positive trace products are established using Mathlib's PSD square root and congruence theorem. No nonnegativity of the desired remainder is assumed. Section 7 supplies particular 2×2 matrices and derives the optical interpretation by an ordinary proof. Those substitutions and the full joint envelope remain outside the formal certificate's scope.

B.3 Important statements and their exact formal scope

Manuscript statement	Declaration or status	Formal scope
Complex reciprocal two-port	`twoPort_reciprocal`; `twoPort_determinant`	Actual 2×2 complex matrix and its determinant.
Absorption determinant	`twoPort_absorption_determinant`	Actual determinant of $I - S^\dagger S$; not a Maxwell flux identity.
Eliminated envelope	`envelope_identity`; `determinant_budget_iff_shifted_square`	Real algebra under an explicit quadratic identity. Does not prove all of the matrix-feasibility theorem.
Phase penalty and inactive branch	`phase_penalty_identity`; `phase_penalty_nonnegative`; `inactive_branch_identity`	Polynomial identities and elementary nonnegativity.
Rational target tangent and gap	`target_circle_identity`; `target_envelope_tangent`; `target_gap_from_det_floor`; `explicit_depth_constant`	The determinant floor is a hypothesis. The parameters are the stated rational benchmark, not arbitrary material profiles.
Lemma 4.1, complex comparison	**Unverified**	No formal Helmholtz initial-value problem or modulus comparison.
Lemma 4.2, capped Fourier moment	**Unverified**	No formal density, integral, or capped-moment optimization.
Lemma 4.3, cell determinants	**Unverified**	The Gram–Wronskian integral, PSD Minkowski inequality, and cell partition are not formalized here.

Manuscript statement	Declaration or status	Formal scope
Lemma 4.4, convexity and tangent	`root_convexity_numerator`; `root_convexity_derivative_algebra`	Only two algebraic cancellations. They do not prove derivatives, convexity, or the global tangent.
Theorem 4.5, resource floor	**Unverified**	No material-constrained Maxwell class or uniform lower-limit theorem.
Section 5, construction and exact matching	**Unverified**	No matrix-exponential remainder, physical layer realization, or implicit-function proof.
Theorem 5.1	**Unverified; declaration unavailable**	No `OQ7.theorem51` was found or synthesized from assumed bounds.
Single passive output bound	`absorbedPower_nonneg`; `extinctionPower_nonneg`; `extinction_sq_le_four_scattering`; `absorption_le_extinction_parabola`	Normed additive group; actual norms, a unit reference, and output norm at most one. No physical scattering operator is assumed.
Equal-extinction pair inequality	`equal_extinction_absorption_asymmetry`	Actual norm-defined powers; equal extinction is an explicit premise.
Finite-port optical identity	`column_extinction_formula`; `reciprocal_column_extinction`	The first is derived by expanding complex norm squares; the second derives extinction symmetry from matrix transpose symmetry.
Passive column absorption	`column_absorption_eq_diagonal`; `passive_column_absorption_nonneg`	Mathlib complex positive semidefiniteness applied to a unit port vector.
Finite-port parabola and pair bound	`column_extinction_sq_le_four_scattering`; `column_absorption_le_extinction_parabola`; `finite_port_absorption_asymmetry`	Complete inequalities for the above column/reference definitions.
Real witness in (48)	`threePort_reciprocal`; `threePort_loss_factorization`; `threePort_loss_quadratic`; `threePort_loss_nonnegative`; `threePort_sharp_value`	Real matrix identities. The complex extension below separately proves full complex passivity.
Complex witness and passivity	`threePortComplex_reciprocal`; `threePortComplex_loss_gram`; `threePortComplex_passive`	Exact complex Gram factorization, hence genuine complex PSD.
Witness observables and equality	`threePortComplex_absorption_left`; `threePortComplex_absorption_right`; `threePortComplex_extinction`; `threePortComplex_saturates`	Computed absorption and extinction under $t^2 + s^2 = 1$.
Existence at every $X \in [0, 4]$	`finite_port_bound_attained`	Explicit complex matrix existence, with no Maxwell-realizability claim.
Positive trace products	`trace_nonneg_of_posSemidef`; `trace_product_nonneg_of_posSemidef`	Actual finite real PSD matrices; the product trace is nonnegative even though the product need not be symmetric.
Joint-bound supporting certificate	`overlap_certificate_identity`; `overlap_psd_trace_certificate`	Exact three-trace identity and inequality under symmetry, PSD budgets, and the support identity. The certificate is proved in every finite dimension.

Manuscript statement	Declaration or status	Formal scope
Theorem 7.1, full joint envelope	**Ordinary proof; certificate formalized**	The canonical optical compression, parameterized certificate matrices, sharp values, and equality operators are not formally linked to the generic certificate.
Scalar fixed-extinction and golden-ratio optimum	**Ordinary proof only**	Theorem 6.3 is not among the formal declarations.
Vector dielectric and finite-power beams	**Ordinary proof only**	Theorem 6.4 and Corollary 6.5 require physical and analytic correspondence not present in the Lean model.

B.4 Reproducibility and preservation record

The source package includes the original PDF and original reproducibility archive without modification, a separate revised manuscript, and a hash ledger. The original numerical scripts run under Python 3.12.14, NumPy 2.3.5, SciPy 1.17.0, and Matplotlib 3.10.8. Exact polynomial checks also use SymPy 1.13.3. The four reproduced JSON files are byte-identical to their originals; the comparison is recorded in `reproduction/comparison_with_originals.json`.

The new experiments provide three independently written normal-incidence implementations: a matrix exponential calculation, a closed scattering cascade, and a modal interface-continuity solve. The beam check uses the oblique four-component Maxwell equations. Their outputs, random seeds, optimization tolerances, and assertions remain in the scripts and JSON records. Numerical calculations diagnose errors; none supplies an omitted analytic or formal proof.

From the extracted source root, the entry points are

```
python3 scripts/reproduce_numerics.py
cd lean
bash scripts/bootstrap-linux.sh
cd ../manuscript
latexmk -pdf -interaction=nonstopmode -halt-on-error oq7_generalized.tex
```

The top-level README states dependency installation, alternative explicit build commands, file layout, and the exact remaining formalization blockers. Build logs document both successful checks and relevant failed approaches; a previous failed attempt is not represented as the final status.

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